MITOCHONDRIA · ASTRONAUT HEALTH INTELLIGENCE

Health Intelligence Report

Preliminary automated assessment — for Ground Medical review. Not a physician's diagnosis.

ASTRONAUT	MISSION	ASSESSMENT PERIOD	EXAMINATIONS WITH DATA
AI	MITO_DEMO_2026	2026-11-01 to 2026-11-30	3, 6, 9, 12, 16, 20, 24, 28 Nov 2026 (8 dates)

SYNTHETIC DEMONSTRATION DATA — not real astronaut data. Age, sex, symptoms, medications and history are not recorded or assumed.

LITERATURE-BASED SYNTHETIC DEMONSTRATION — NOT CLINICALLY VALIDATED

Literature criteria and synthetic identity mappings colour results for this demonstration only: not Researcher-approved, never validated alerts, never applied to real data.

Most important findings

HIGH PRIORITY MEDICAL REVIEW

Clinical class AMBER (PROVISIONAL)

Fever and inflammation with abnormal vital signs

Oxygen saturation 91 %; Systolic blood pressure 88 mmHg; temperature 38.8 °C; temperature 38.2 °C; hsCRP 28 mg/L; hsCRP 22 mg/L; hsCRP 12 mg/L; IL-6 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL; lactate 4.2 mmol/L; lactate 3.0 mmol/L (12 Nov, 16 Nov).

Why this priority: PP-1: Oxygen saturation (SpO2), Systolic blood pressure in a NEWS2 3-point band. PP-2: fever and CRP above 10 mg/L on the same or adjacent dates.

Next: Ground Medical clinical assessment now: symptoms, mentation and a complete repeated set of vital signs.

HIGH PRIORITY MEDICAL REVIEW

Clinical class NEUTRAL

Raised D-dimer; fibrinogen without an applicable criterion

D-dimer 0.82 mg/L FEU; D-dimer 2.8 mg/L FEU; D-dimer 2.1 mg/L FEU; D-dimer 1.2 mg/L FEU; D-dimer 0.69 mg/L FEU; fibrinogen 295 mg/dL; fibrinogen 320 mg/dL; fibrinogen 390 mg/dL; fibrinogen 520 mg/dL; fibrinogen 485 mg/dL; fibrinogen 420 mg/dL; fibrinogen 350 mg/dL; fibrinogen 305 mg/dL, with Oxygen saturation 91 % and Oxygen saturation 93 % …

Why this priority: PP-2: D-dimer above the cut-off converges with SpO2 or respiratory rate in a NEWS2 ≥2-point band within one day.

Next: Ground Medical to assess the clinical probability of venous thromboembolism before any further testing.

DATA CHECK

Clinical class NEUTRAL DATA CHECK

HbA1c results need verification

HbA1c 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 % (3 Nov, 6 Nov, 9 Nov, 12 Nov, 16 Nov, 20 Nov, 24 Nov, 28 Nov) — held for measurement verification.

Why this priority: PP-5: every HbA1c result is held by the existing measurement-consistency rule; no clinical statement is made until it is repeated.

Next: Repeat HbA1c on a new sample with the method recorded, plus fasting glucose.

MONITORING OR FURTHER EVALUATION

4 further findings

- Electrolyte shifts with a prolonged QTc — monitoring + pattern data check
- Raised triglycerides with an unfavourable cholesterol profile — monitoring
- Bone-turnover change and low vitamin D — monitoring
- Poor sleep efficiency and reduced heart-rate variability — monitoring

Next: See the Preliminary Medical Assessment.

RED	High priority medical review; not a diagnosis	AMBER	Monitoring or further evaluation	DATA CHECK	Purple — verify the measurement first
GREEN	Within applicable reference criteria	NEUTRAL	Insufficient classification evidence	(PROVISIONAL)	Literature-based class; not clinically validated
TREND REVIEW	Significant change between examinations; not a priority				

Clinical classes: 0 Researcher-approved criteria for this mission; PROVISIONAL = documented literature criterion (synthetic demonstration only; never a validated alert); 5 synthetic identity mapping(s) applied (owner-directed). Panel colour = warning priority, separate from each result's clinical class. General-population criteria are not NASA flight limits.

Priority next steps

1. Ground Medical clinical assessment now: symptoms, mentation and a complete repeated set of vital signs.

Executive health overview

8 examination date(s) with 248 observation(s) in 31 biomarker series. 7 body system(s) show flagged or classified results; 17 series show a significant change between examinations (trend review) and 1 require measurement verification first. Medical-review priority: 2 finding(s) high priority, 4 monitoring, 1 data check.

1 · What is happening?
7 connected finding pattern(s) for Ground Medical review: fever and inflammation with abnormal vital signs (high priority medical review); raised d-dimer; fibrinogen without an applicable criterion (high priority medical review); hba1c results need verification (data check); electrolyte shifts with a prolonged qtc (monitoring or further evaluation); raised triglycerides with an unfavourable cholesterol profile (monitoring or further evaluation); bone-turnover change and low vitamin d (monitoring or further evaluation); poor sleep efficiency and reduced heart-rate variability (monitoring or further evaluation).

2 · Which results are abnormal or concerning?
Classified: Potassium 5.8 / 3.2 mmol/L (AMBER, provisional); Sodium 128 / 131 / 146 mmol/L (AMBER, provisional); HDL cholesterol 39 mg/dL (AMBER, provisional); Triglycerides 155 / 165 / 230 / 330 / 288 / 216 / 184 / 160 mg/dL (AMBER, provisional); hsCRP (inflammation) 28 / 22 / 12 mg/L (AMBER, provisional); Diastolic blood pressure 54 / 58 mmHg (AMBER, provisional); Systolic blood pressure 88 mmHg (AMBER, provisional); Vitamin D (25-hydroxy) 12 / 12 / 12 / 12 / 13 / 13 / 14 / 14 ng/mL (AMBER, provisional); Sleep efficiency 69 / 52 / 56 / 66 % (AMBER, provisional). Trend review — significant change between examinations (not a medical-review priority or a classification): Potassium, Fasting glucose, HbA1c (3-month average blood sugar), Triglycerides, hsCRP (inflammation), IL-10 (anti-inflammatory signal), IL-6 (inflammatory signal), TNF-α, D-dimer, Fibrinogen, Lactate, Resting heart rate, CTX (bone breakdown), Osteocalcin (bone formation), Heart-rate variability (RMSSD), Respiratory rate, White blood cell count.

3 · What increased or decreased?
Increased: Fasting glucose 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL (+2.1 %) (intermediate peak 172); LDL cholesterol 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL (+2.5 %) (intermediate peak 145); Triglycerides 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL (+3.2 %) (intermediate peak 330); hsCRP (inflammation) 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L (+75.0 %) (intermediate peak 28); IL-10 (anti-inflammatory signal) 4 → 5 → 8 → 15 → 13 → 9 → 6 → 5 pg/mL (+25.0 %) (intermediate peak 15); IL-6 (inflammatory signal) 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL (+39.7 %) (intermediate peak 74); TNF-α 8 → 10 → 15 → 24 → 20 → 15 → 11 → 9 pg/mL (+12.5 %) (intermediate peak 24); D-dimer 0.32 → 0.41 → 0.82 → 2.8 → 2.1 → 1.2 → 0.69 → 0.39 mg/L FEU (+21.9 %) (intermediate peak 2.8); Fibrinogen 295 → 320 → 390 → 520 → 485 → 420 → 350 → 305 mg/dL (+3.4 %) (intermediate peak 520); Lactate 1.1 → 1.3 → 1.8 → 4.2 → 3.0 → 1.9 → 1.4 → 1.2 mmol/L (+9.1 %) (intermediate …

4 · Which changes are biologically meaningful?
Plausible and important if confirmed: Potassium, Fasting glucose, Triglycerides, hsCRP (inflammation), IL-10 (anti-inflammatory signal), IL-6 (inflammatory signal), TNF-α, D-dimer, Fibrinogen, Lactate, Resting heart rate, CTX (bone breakdown), Osteocalcin (bone formation), Heart-rate variability (RMSSD), Respiratory rate, White blood cell count. Verify the measurement first: HbA1c (3-month average blood sugar).

5 · Which body systems are affected?
Fluids & electrolytes; Glucose & lipids; Inflammation & immunity; Blood & clotting; Heart & circulation; Bone & mineral; Sleep & recovery

6 · How are the biomarkers related?
Fever and inflammation with abnormal vital signs – Raised triglycerides with an unfavourable cholesterol profile: Inflammation raises plasma triglycerides (established; Khovidhunkit 2004); Fever and inflammation with abnormal vital signs – Raised D-dimer; fibrinogen without an applicable criterion: Acute-phase response raises fibrinogen; D-dimer is non-specific; also raised D-dimer with low SpO2 / fast breathing (established; Gabay 1999); Fever and inflammation with abnormal vital signs – Poor sleep efficiency and reduced heart-rate variability: Illness may reduce sleep quality and HRV (hypothesis); Fever and inflammation with abnormal vital signs – Electrolyte shifts with a prolonged QTc: Same dates — co-occurrence, not cause. Co-occurrence does not show cause.

7 · Which conditions could explain the findings?

acute infection with a systemic inflammatory response (no source is recorded).; non-infectious inflammation, such as tissue injury or strenuous exertion.; a measurement or transcription error affecting several vital signs (less likely if they come from independent devices; devices are not recorded).; acute-phase response to the inflammation recorded in the same period.; venous thromboembolism — to be assessed clinically; not indicated by the measurements alone.; sample mix-up or transcription error.. All are hypotheses for clinical evaluation, not diagnoses.

8 · What supports or contradicts them?

Fever and inflammation with abnormal vital signs: 17 supporting, 11 contradicting/limiting; Raised D-dimer; fibrinogen without an applicable criterion: 17 supporting, 1 contradicting/limiting; HbA1c results need verification: 1 supporting, 0 contradicting/limiting; Electrolyte shifts with a prolonged QTc: 3 supporting, 1 contradicting/limiting; Raised triglycerides with an unfavourable cholesterol profile: 2 supporting, 1 contradicting/limiting; Bone-turnover change and low vitamin D: 11 supporting, 1 contradicting/limiting; Poor sleep efficiency and reduced heart-rate variability: 5 supporting, 0 contradicting/limiting.

9 · What should the astronaut do?

Report any symptoms to Ground Medical now (for example fever, breathlessness, chest pain, palpitations or dizziness). Do not change fluid, salt or potassium intake before Ground Medical review. Do not start or change supplements or exercise until Ground Medical has reviewed these findings.

10 · What should Ground Medical confirm?

Ground Medical to assess the crew member clinically: symptoms, mentation and a full repeated set of vital signs at the same time. Ground Medical to estimate the clinical probability of venous thromboembolism with a validated score before deciding on further testing; imaging only if the clinician judges it indicated. Repeat HbA1c on a new sample with the method recorded, together with a fasting plasma glucose, if Ground Medical agrees.

Body systems at a glance

Fluids & electrolytes

AMBER (PROVISIONAL) TREND REVIEW

Potassium 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L; Sodium 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L

Glucose & lipids

AMBER (PROVISIONAL) TREND REVIEW DATA CHECK

Fasting glucose 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL; HbA1c (3-month average blood sugar) 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 %; HDL cholesterol 48 → 46 → 43 → 39 → …

Inflammation & immunity

AMBER (PROVISIONAL) TREND REVIEW

hsCRP (inflammation) 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L; IL-10 (anti-inflammatory signal) 4 → 5 → 8 → 15 → 13 → 9 → 6 → 5 pg/mL; IL-6 (inflammatory signal) 5.8 → 12 → 32 → …

Blood & clotting

TREND REVIEW

D-dimer 0.32 → 0.41 → 0.82 → 2.8 → 2.1 → 1.2 → 0.69 → 0.39 mg/L FEU; Fibrinogen 295 → 320 → 390 → 520 → 485 → 420 → 350 → 305 mg/dL; aPTT (clotting time) 29 → 30 → 31 → 34 → 32 …

Heart & circulation

AMBER (PROVISIONAL) TREND REVIEW

Diastolic blood pressure 76 → 73 → 69 → 54 → 58 → 65 → 72 → 75 mmHg; Lactate 1.1 → 1.3 → 1.8 → 4.2 → 3.0 → 1.9 → 1.4 → 1.2 mmol/L; Resting heart rate 64 → 69 → 77 → 108 → 98 → 83 → 72 → 66 …

Bone & mineral

AMBER (PROVISIONAL) TREND REVIEW

CTX (bone breakdown) 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL; Osteocalcin (bone formation) 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL; Vitamin D (25-hydroxy) 12 → 12 …

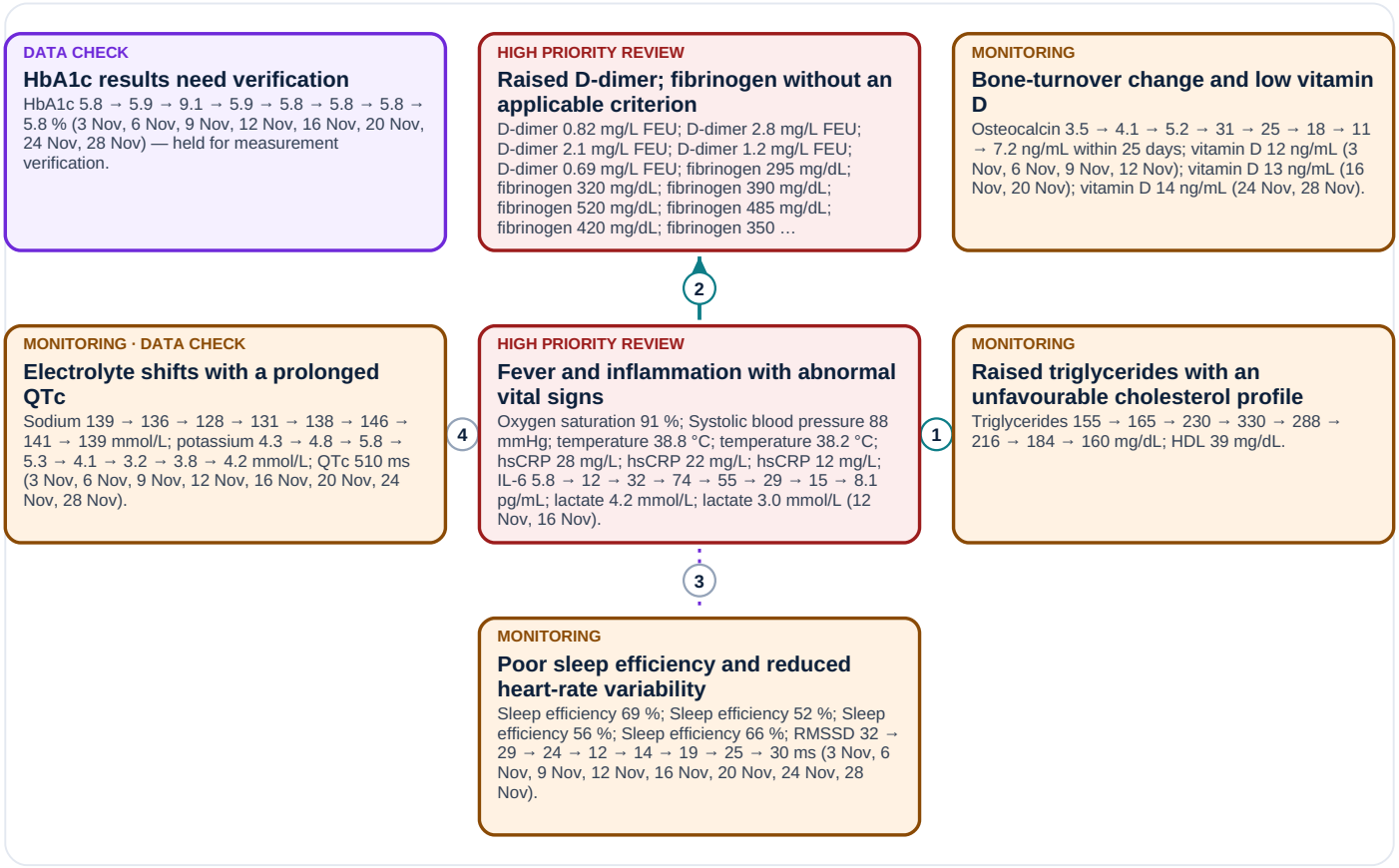
Sleep & recovery

AMBER (PROVISIONAL) TREND REVIEW

Heart-rate variability (RMSSD) 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms; Sleep efficiency 83 → 78 → 69 → 52 → 56 → 66 → 75 → 81 %

Integrated health assessment

Findings are read as connected patterns. Box colour = warning priority. Solid teal arrows: established biological links with a source. Dashed grey: co-occurrence in this period (not cause). Dotted violet: hypothesis. Numbers refer to the legend below the map.



1. Fever and inflammation with abnormal vital signs — Raised triglycerides with an unfavourable cholesterol profile. Established: Inflammation raises plasma triglycerides (Khovidhunkit 2004).
2. Fever and inflammation with abnormal vital signs — Raised D-dimer; fibrinogen without an applicable criterion. Established: Acute-phase response raises fibrinogen; D-dimer is non-specific; also raised D-dimer with low SpO2 / fast breathing (Gabay 1999).
3. Fever and inflammation with abnormal vital signs — Poor sleep efficiency and reduced heart-rate variability. Hypothesis: Illness may reduce sleep quality and HRV.
4. Fever and inflammation with abnormal vital signs — Electrolyte shifts with a prolonged QTc. Co-occurrence: Same dates — co-occurrence, not cause.

Overall picture

2 pattern(s) warrant high-priority Ground Medical review; the leading one is fever and inflammation with abnormal vital signs. Part of the picture may follow from it through established biology: raised triglycerides with an unfavourable cholesterol profile (inflammation raises plasma triglycerides; Khovidhunkit 2004); raised D-dimer; fibrinogen without an applicable criterion (acute-phase response raises fibrinogen; D-dimer is non-specific; also raised D-dimer with low SpO2 / fast breathing; Gabay 1999). Other findings occur at the same time without an established causal link: poor sleep efficiency and reduced heart-rate variability; electrolyte shifts with a prolonged QTc — co-occurrence does not show cause.

Before interpretation, measurements must be verified for: HbA1c results need verification; electrolyte shifts with a prolonged QTc. A data check never lowers a medical-review priority.

Independent findings for monitoring: bone-turnover change and low vitamin D.

All patterns are preliminary, rest on synthetic demonstration data and literature thresholds that are not Researcher-approved, and require clinical evaluation; none is a diagnosis.

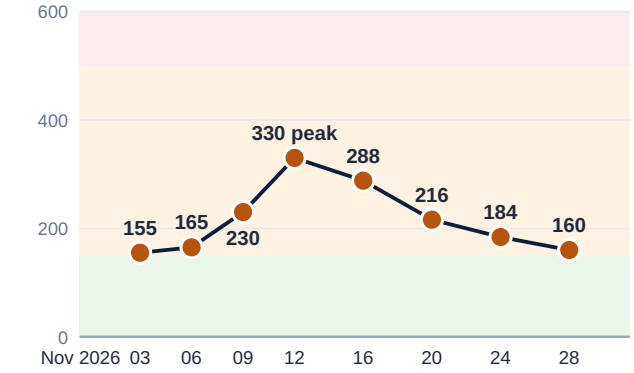
Longitudinal biomarker intelligence

Every chart shows only observed values, on their actual dates and in their original units. Coloured bands appear only where an approved (or, in demonstration mode, a provisional literature) criterion applies. Hollow violet markers and dashed lines: measurement must be verified first. Grey: unclassified.

Triglycerides

mg/dL

MONITORING | TREND REVIEW



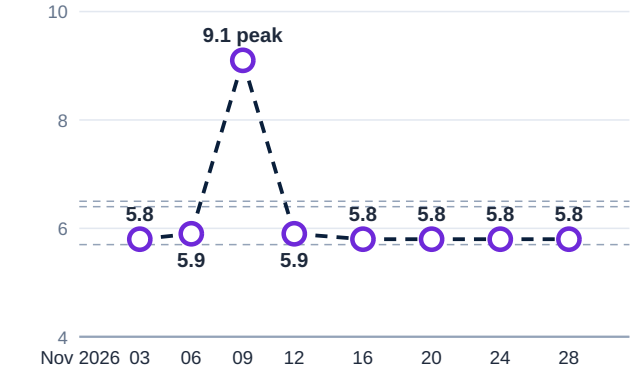
Healthy (NHLBI, adults): <150 mg/dL
Borderline high (NHLBI): 150–199 mg/dL
High (NHLBI): 200–499 mg/dL
Very high (NHLBI): >500 mg/dL

Triglycerides. 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 330 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +5 mg/dL (+3.2%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

HbA1c (3-month average blood sugar)

%

DATA CHECK | TREND REVIEW



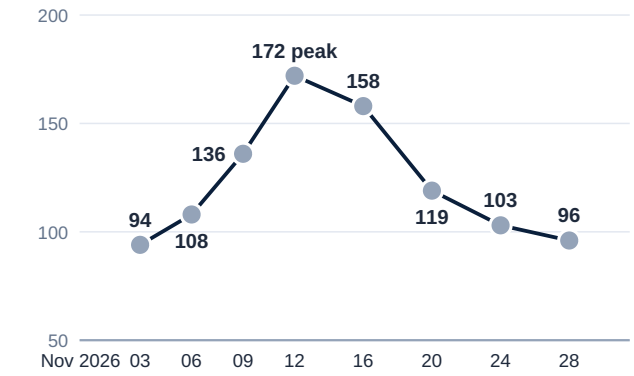
Dashed lines at 5.7 / 6.4 / 6.5 % — orientation only, not a criterion

HbA1c (3-month average blood sugar). 5.8 → 5.9 → 9.1 → 5.9 → 5.8 → 5.8 → 5.8 → 5.8 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 9.1 % on 2026-11-09 (intermediate maximum), then fell again; first to last 0 % (0%). 5.9 → 9.1 % (+54.2%) from 2026-11-06 to 2026-11-09: a 3-day interval is far shorter than the period HbA1c reflects; measurement consistency must be checked before interpretation.

Fasting glucose

mg/dL

TREND REVIEW



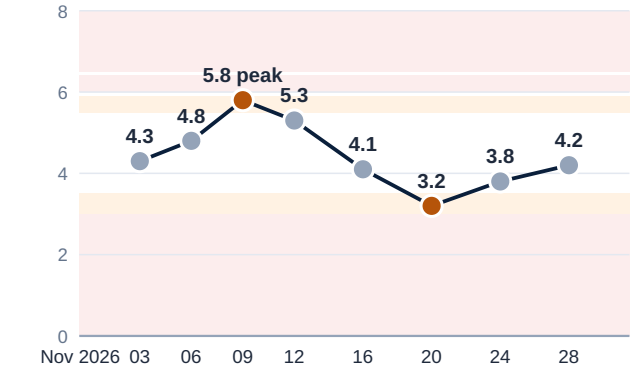
Fasting glucose. 94 → 108 → 136 → 172 → 158 → 119 → 103 → 96 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 172 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +2 mg/dL (+2.1%). No approved reference criterion applies, so no reference band is drawn.

Potassium

mmol/L

MONITORING | TREND REVIEW

+ pattern data check † — verify this series before interpretation



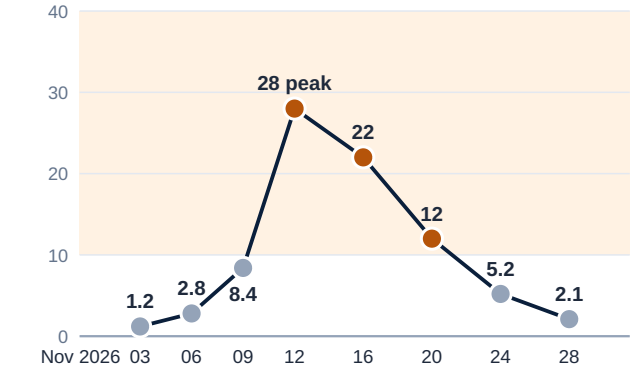
Mild hyperkalaemia (UKKA): 5.5–5.9 mmol/L
Moderate hyperkalaemia (UKKA): 6–6.4 mmol/L
Severe hyperkalaemia (UKKA): ≥6.5 mmol/L
Mild hypokalaemia (StatPearls): ≥3 and <3.5 mmol/L
Moderate hypokalaemia (StatPearls): ≥2.5 and <3 mmol/L
Severe hypokalaemia (StatPearls): <2.5 mmol/L

Potassium. 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 5.8 mmol/L on 2026-11-09 (intermediate maximum), then fell again; first to last -0.1 mmol/L (-2.3%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

hsCRP (inflammation)

mg/L

Associated with HIGH PRIORITY pattern | TREND REVIEW

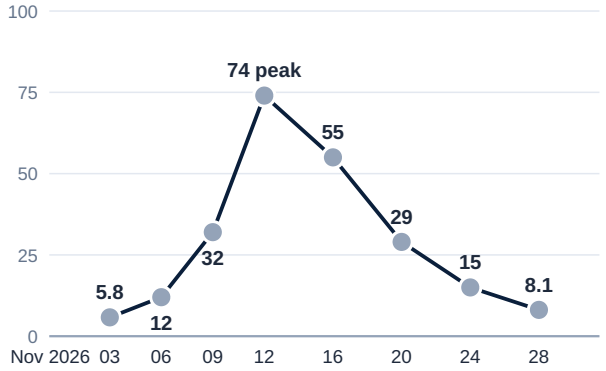


hsCRP (inflammation). 1.2 → 2.8 → 8.4 → 28 → 22 → 12 → 5.2 → 2.1 mg/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 28 mg/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.9 mg/L (+75%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

IL-6 (inflammatory signal)

pg/mL

Associated with HIGH PRIORITY pattern | TREND REVIEW

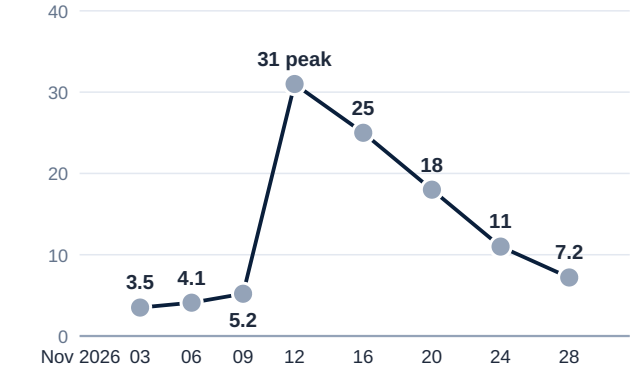


IL-6 (inflammatory signal). 5.8 → 12 → 32 → 74 → 55 → 29 → 15 → 8.1 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 74 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +2.3 pg/mL (+39.7%). No approved reference criterion applies, so no reference band is drawn.

Osteocalcin (bone formation)

ng/mL

MONITORING | TREND REVIEW

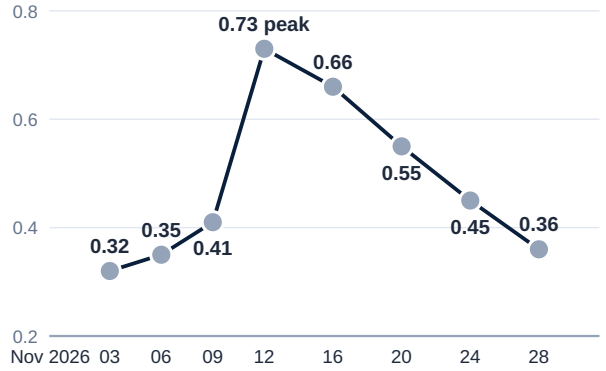


Osteocalcin (bone formation). 3.5 → 4.1 → 5.2 → 31 → 25 → 18 → 11 → 7.2 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 31 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3.7 ng/mL (+105.7%). No approved reference criterion applies, so no reference band is drawn.

CTX (bone breakdown)

ng/mL

MONITORING | TREND REVIEW

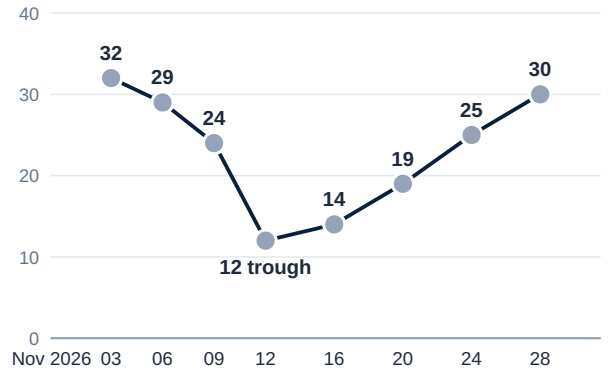


CTX (bone breakdown). 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 0.73 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +0.04 ng/mL (+12.5%). No approved reference criterion applies, so no reference band is drawn.

Heart-rate variability (RMSSD)

ms

MONITORING | TREND REVIEW

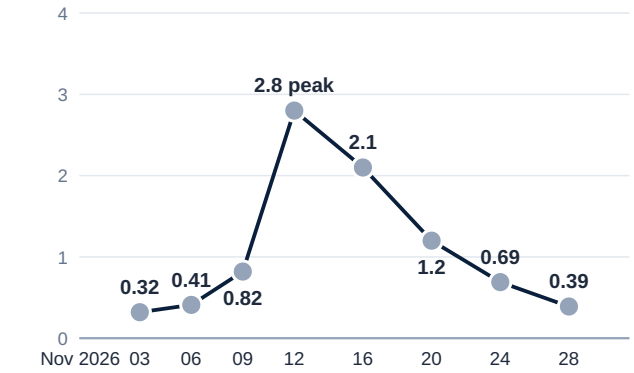


Heart-rate variability (RMSSD). 32 → 29 → 24 → 12 → 14 → 19 → 25 → 30 ms from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 12 ms on 2026-11-12 (intermediate minimum), then rose again; first to last -2 ms (-6.3%). No approved reference criterion applies, so no reference band is drawn.

D-dimer

mg/L FEU

Associated with HIGH PRIORITY pattern | TREND REVIEW

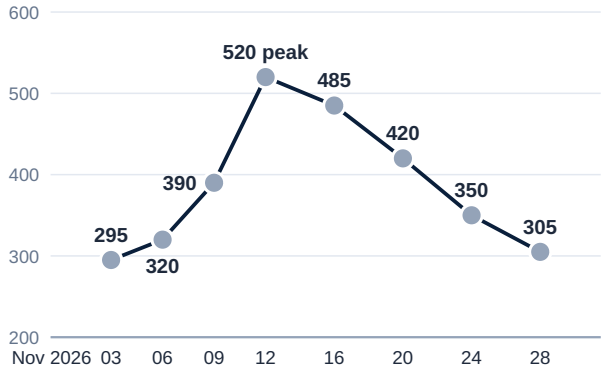


D-dimer. 0.32 → 0.41 → 0.82 → 2.8 → 2.1 → 1.2 → 0.69 → 0.39 mg/L FEU from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 2.8 mg/L FEU on 2026-11-12 (intermediate maximum), then fell again; first to last +0.07 mg/L FEU (+21.9%). No approved reference criterion applies, so no reference band is drawn.

Fibrinogen

mg/dL

Associated with HIGH PRIORITY pattern | TREND REVIEW

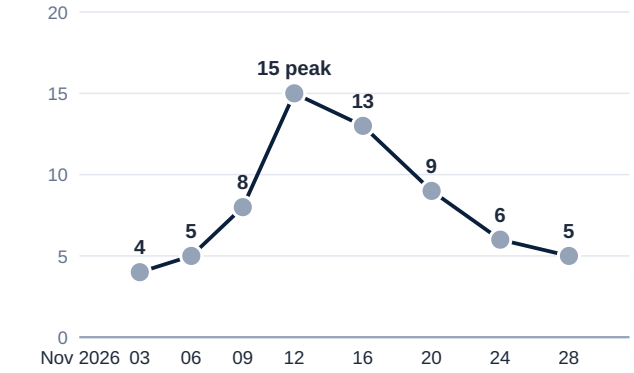


Fibrinogen. 295 → 320 → 390 → 520 → 485 → 420 → 350 → 305 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 520 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +10 mg/dL (+3.4%). No approved reference criterion applies, so no reference band is drawn.

IL-10 (anti-inflammatory signal)

pg/mL

TREND REVIEW

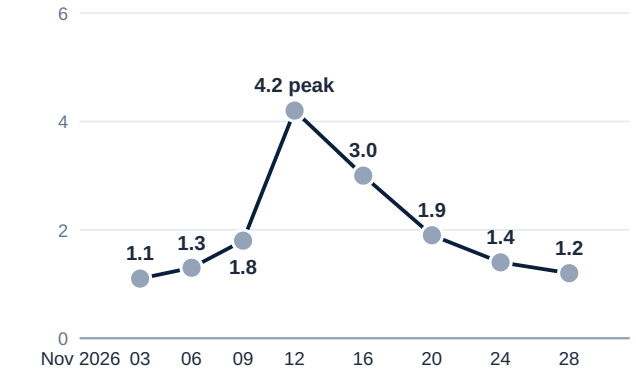


IL-10 (anti-inflammatory signal). 4 → 5 → 8 → 15 → 13 → 9 → 6 → 5 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 15 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +1 pg/mL (+25%). No approved reference criterion applies, so no reference band is drawn.

Lactate

mmol/L

Associated with HIGH PRIORITY pattern | TREND REVIEW

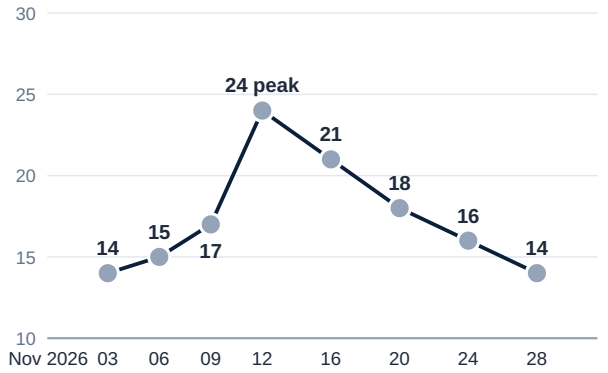


Lactate. 1.1 → 1.3 → 1.8 → 4.2 → 3.0 → 1.9 → 1.4 → 1.2 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 4.2 mmol/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.1 mmol/L (+9.1%). No approved reference criterion applies, so no reference band is drawn.

Respiratory rate

breaths/min

Associated with HIGH PRIORITY pattern | TREND REVIEW

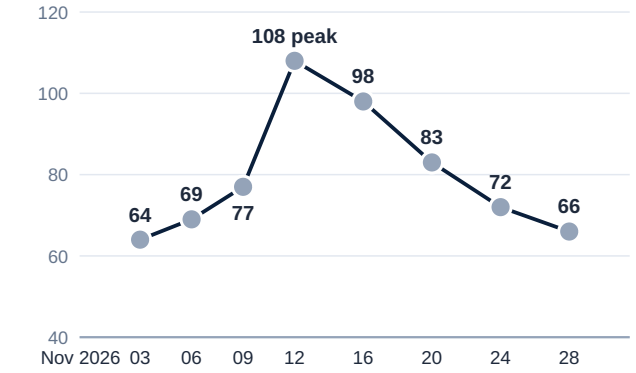


Respiratory rate. 14 → 15 → 17 → 24 → 21 → 18 → 16 → 14 breaths/min from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 24 breaths/min on 2026-11-12 (intermediate maximum), then fell again; first to last 0 breaths/min (0%). No approved reference criterion applies, so no reference band is drawn.

Resting heart rate

bpm

TREND REVIEW

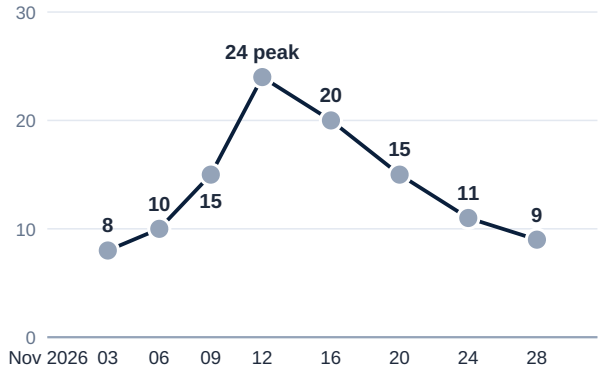


Resting heart rate. 64 → 69 → 77 → 108 → 98 → 83 → 72 → 66 bpm from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 108 bpm on 2026-11-12 (intermediate maximum), then fell again; first to last +2 bpm (+3.1%). No approved reference criterion applies, so no reference band is drawn.

TNF-α

pg/mL

TREND REVIEW

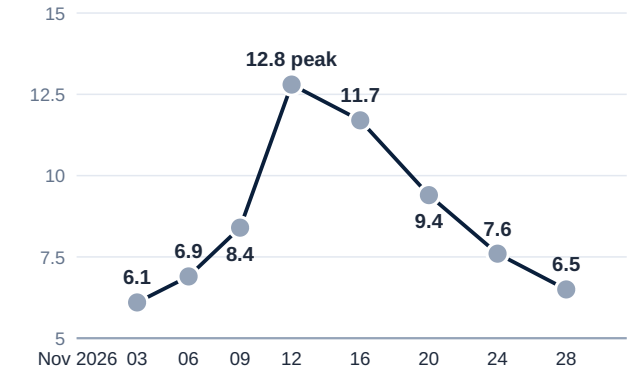


TNF-α. 8 → 10 → 15 → 24 → 20 → 15 → 11 → 9 pg/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 24 pg/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +1 pg/mL (+12.5%). No approved reference criterion applies, so no reference band is drawn.

White blood cell count

10^9/L

TREND REVIEW

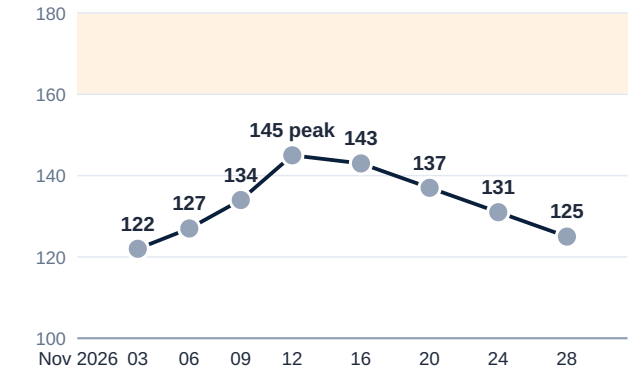


White blood cell count. 6.1 → 6.9 → 8.4 → 12.8 → 11.7 → 9.4 → 7.6 → 6.5 10^9/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 12.8 10^9/L on 2026-11-12 (intermediate maximum), then fell again; first to last +0.4 10^9/L (+6.6%). No approved reference criterion applies, so no reference band is drawn.

LDL cholesterol

mg/dL

MONITORING



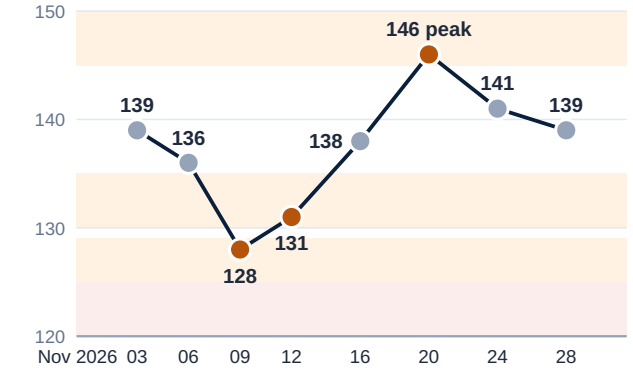
High (ATP III): 160–189 mg/dL
Very high (ATP III): ≥190 mg/dL

LDL cholesterol. 122 → 127 → 134 → 145 → 143 → 137 → 131 → 125 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 145 mg/dL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 mg/dL (+2.5%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

Sodium

mmol/L

MONITORING



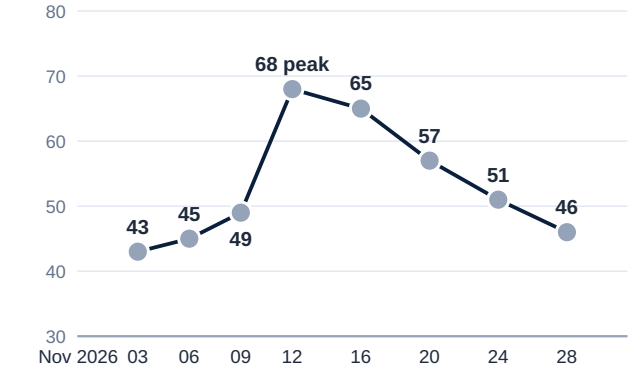
Mild hyponatraemia (biochemical, ESE): ≥130 and <135 mmol/L
Moderate hyponatraemia (biochemical, ESE): 125–129 mmol/L
Profound hyponatraemia (biochemical, ESE): <125 mmol/L
Hypernatraemia (StatPearls): >145 mmol/L

Sodium. 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 146 mmol/L on 2026-11-20 (intermediate maximum), then fell again; first to last 0 mmol/L (0%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

P1NP (bone formation)

ng/mL

MONITORING

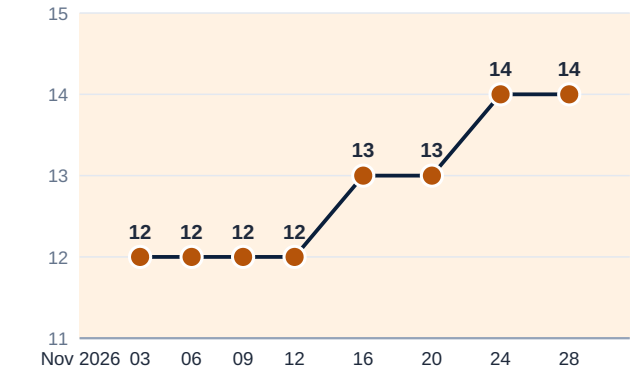


P1NP (bone formation). 43 → 45 → 49 → 68 → 65 → 57 → 51 → 46 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 68 ng/mL on 2026-11-12 (intermediate maximum), then fell again; first to last +3 ng/mL (+7%). No approved reference criterion applies, so no reference band is drawn.

Vitamin D (25-hydroxy)

ng/mL

MONITORING



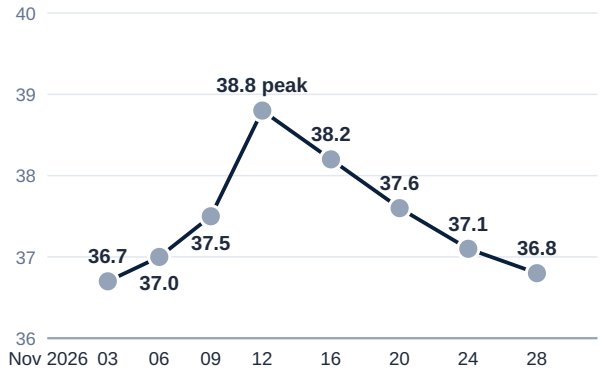
Associated with deficiency (NIH ODS): <12 ng/mL
Generally considered inadequate (NIH ODS): ≥12 and <20 ng/mL
Generally considered adequate (NIH ODS): 20–50 ng/mL
Linked to potential adverse effects (NIH ODS): >50 ng/mL

Vitamin D (25-hydroxy). 12 → 12 → 12 → 12 → 13 → 13 → 14 → 14 ng/mL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Rose at every examination; first to last +2 ng/mL (+16.7%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

Body temperature

°C

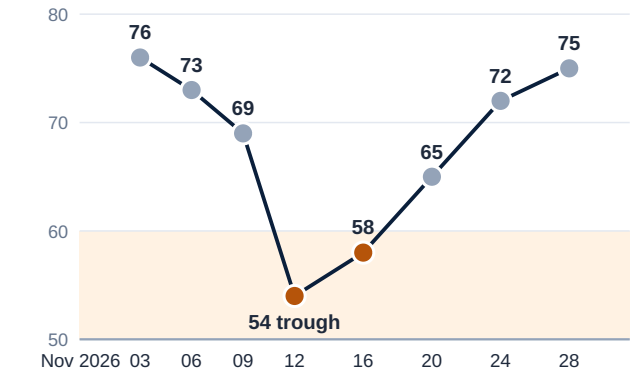
Associated with HIGH PRIORITY pattern



Body temperature. 36.7 → 37.0 → 37.5 → 38.8 → 38.2 → 37.6 → 37.1 → 36.8 °C from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 38.8 °C on 2026-11-12 (intermediate maximum), then fell again; first to last +0.1 °C (+0.3%). No approved reference criterion applies, so no reference band is drawn.

Diastolic blood pressure

mmHg



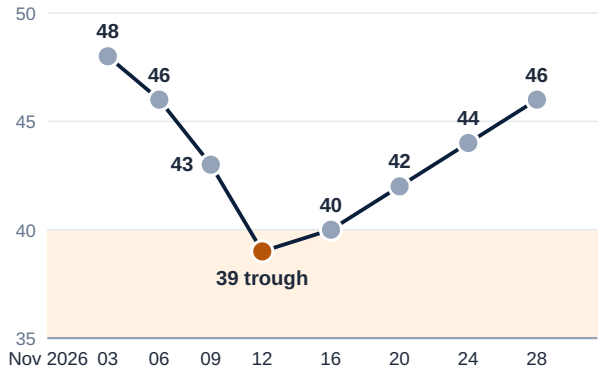
Below 60 mmHg (NHLBI low blood pressure): <60 mmHg

Diastolic blood pressure. 76 → 73 → 69 → 54 → 58 → 65 → 72 → 75 mmHg from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 54 mmHg on 2026-11-12 (intermediate minimum), then rose again; first to last -1 mmHg (-1.3%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

HDL cholesterol

mg/dL

MONITORING



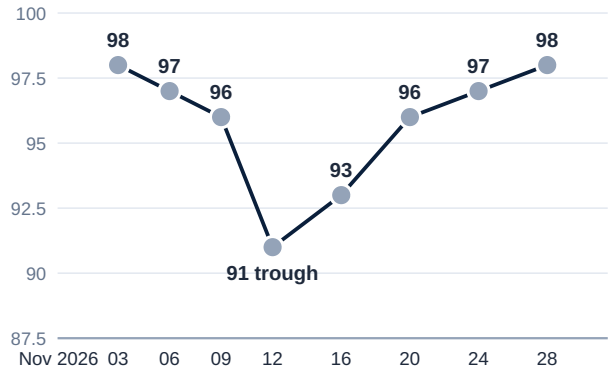
Low HDL (ATP III): <40 mg/dL

HDL cholesterol. 48 → 46 → 43 → 39 → 40 → 42 → 44 → 46 mg/dL from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 39 mg/dL on 2026-11-12 (intermediate minimum), then rose again; first to last -2 mg/dL (-4.2%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

Oxygen saturation (SpO2)

%

Associated with HIGH PRIORITY pattern

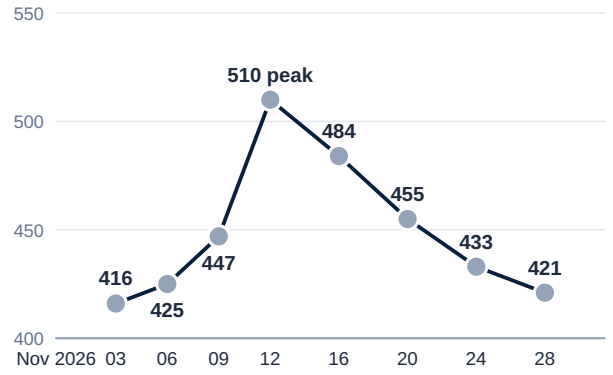


Oxygen saturation (SpO2). 98 → 97 → 96 → 91 → 93 → 96 → 97 → 98 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 91 % on 2026-11-12 (intermediate minimum), then rose again; first to last 0 % (0%). No approved reference criterion applies, so no reference band is drawn.

QTc interval

ms

MONITORING

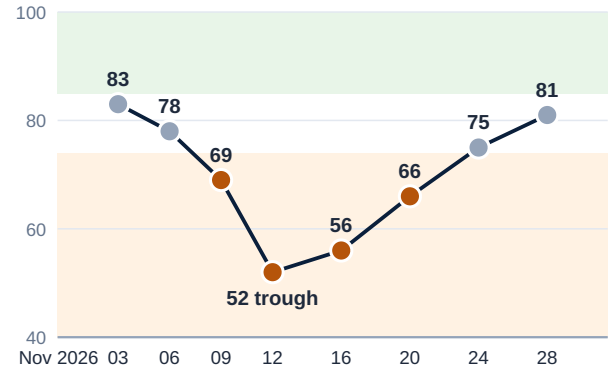


QTc interval. 416 → 425 → 447 → 510 → 484 → 455 → 433 → 421 ms from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Peaked at 510 ms on 2026-11-12 (intermediate maximum), then fell again; first to last +5 ms (+1.2%). No approved reference criterion applies, so no reference band is drawn.

Sleep efficiency

%

MONITORING



≤74 % — does not indicate good sleep quality (NSF): ≤74 %

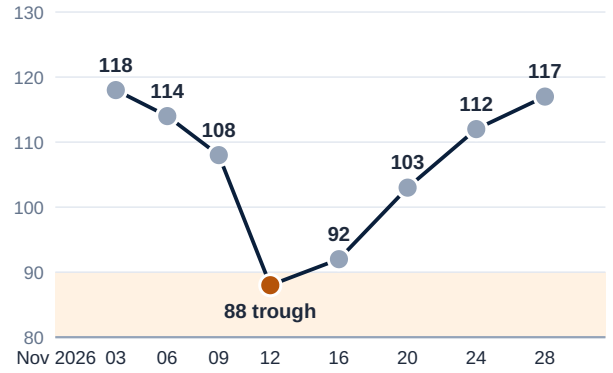
≥85 % — good sleep quality (NSF): ≥85 %

Sleep efficiency. 83 → 78 → 69 → 52 → 56 → 66 → 75 → 81 % from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 52 % on 2026-11-12 (intermediate minimum), then rose again; first to last -2 % (-2.4%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

Systolic blood pressure

mmHg

Associated with HIGH PRIORITY pattern



Below 90 mmHg (NHLBI low blood pressure): <90 mmHg

Systolic blood pressure. 118 → 114 → 108 → 88 → 92 → 103 → 112 → 117 mmHg from 2026-11-03 to 2026-11-28 (8 examinations, 25 days). Fell to 88 mmHg on 2026-11-12 (intermediate minimum), then rose again; first to last -1 mmHg (-0.8%). No Researcher-approved criterion applies; the coloured bands are provisional literature criteria (synthetic demonstration).

Not charted — no classification, review priority or flag in this report; every value is in the complete results table: aPTT (clotting time), Hematocrit, Hemoglobin.

Preliminary Medical Assessment — For Ground Medical Review

Automated preliminary assessment generated from the recorded measurements. It lists possible conditions for a qualified clinician to evaluate; it is not a physician-certified diagnosis and assigns no probabilities. Symptoms, examination findings, medications, age and sex were not available unless shown.

1 · Fever and inflammation with abnormal vital signs

HIGH PRIORITY MEDICAL REVIEW

WARNING PRIORITY	HIGH PRIORITY MEDICAL REVIEW (rule PP-1, PP-2). PP-1: Oxygen saturation (SpO2), Systolic blood pressure in a NEWS2 3-point band. PP-2: fever and CRP above 10 mg/L on the same or adjacent dates.
CLINICAL CLASS	• Systolic blood pressure 88 mmHg (11/12): AMBER (PROVISIONAL) — Below 90 mmHg (NHLBI low blood pressure) • hsCRP (inflammation) 28 mg/L (11/12): AMBER (PROVISIONAL) — Above 10 mg/L — repeat and evaluate (CDC/AHA) • hsCRP (inflammation) 22 mg/L (11/16): AMBER (PROVISIONAL) — Above 10 mg/L — repeat and evaluate (CDC/AHA) • hsCRP (inflammation) 12 mg/L (11/20): AMBER (PROVISIONAL) — Above 10 mg/L — repeat and evaluate (CDC/AHA)
OBSERVED	• Oxygen saturation (SpO2) 91 % on 12 Nov (NEWS2 3-point band; RCP 2017). • Systolic blood pressure 88 mmHg on 12 Nov (NEWS2 3-point band; RCP 2017). • Respiratory rate 24 breaths/min on 12 Nov (NEWS2 2-point band; RCP 2017). • Respiratory rate 21 breaths/min on 16 Nov (NEWS2 2-point band; RCP 2017). • Oxygen saturation (SpO2) 93 % on 16 Nov (NEWS2 2-point band; RCP 2017). • Systolic blood pressure 92 mmHg on 16 Nov (NEWS2 2-point band; RCP 2017). • Body temperature 38.8 °C on 12 Nov — at or above 38 °C, the CDC fever definition. • Body temperature 38.2 °C on 16 Nov — at or above 38 °C, the CDC fever definition. • hsCRP (inflammation) 28 mg/L on 12 Nov — above 10 mg/L; CDC/AHA advise repeating the test and looking for a source of infection or inflammation. • hsCRP (inflammation) 22 mg/L on 16 Nov — above 10 mg/L; CDC/AHA advise repeating the test and looking for a source of infection or inflammation. • hsCRP (inflammation) 12 mg/L on 20 Nov — above 10 mg/L; CDC/AHA advise repeating the test and looking for a source of infection or inflammation. • IL-6 5.8 pg/mL → 12 pg/mL → 32 pg/mL → 74 pg/mL → 55 pg/mL → 29 pg/mL → 15 pg/mL → 8.1 pg/mL (3 Nov → 28 Nov) — significant change (trend review); no criterion classifies IL-6. • Lactate 4.2 mmol/L on 12 Nov — above 2 mmol/L (Sepsis-3, 2016); interpret with the sampling conditions. • Lactate 3.0 mmol/L on 16 Nov — above 2 mmol/L (Sepsis-3, 2016); interpret with the sampling conditions. • SpO2 91 % on 12 Nov is below the 94–98 % target range used for acutely ill adults (BTS 2017). • SpO2 93 % on 16 Nov is below the 94–98 % target range used for acutely ill adults (BTS 2017).
MEASUREMENT QUALITY	• Measurements carry calendar dates only; whether the vital signs were taken at the same time is not established. • Pulse oximetry can read falsely low with poor perfusion, motion or cold extremities; the device and probe site are not recorded. • Temperature measurement site and method are not recorded. • Lactate rises with exertion, tourniquet time and delayed analysis; sampling conditions are not recorded.
CLINICAL PATTERN	Fever, raised inflammatory markers, vital signs in NEWS2 warning bands and a raised lactate were recorded together on 12 Nov and 16 Nov. Together they describe an acute systemic physiological stress pattern that warrants prompt clinical assessment, including assessment for infection. This is a pattern for review, not a diagnosis.
HYPOTHESES	• Acute infection with a systemic inflammatory response (no source is recorded). • Non-infectious inflammation, such as tissue injury or strenuous exertion. • A measurement or transcription error affecting several vital signs (less likely if they come from independent devices; devices are not recorded).
ALSO SUPPORTING	• 12 Nov: respiratory rate 24 breaths/min and systolic pressure 88 mmHg meet 2 of the 3 qSOFA criteria (mentation not recorded) (Sepsis-3, 2016).
CONTRADICTING / LIMITING	• Respiratory rate 18 breaths/min on 20 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Respiratory rate 16 breaths/min on 24 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Respiratory rate 14 breaths/min on 28 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Oxygen saturation (SpO2) 93 % on 16 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Oxygen saturation (SpO2) 96 % on 20 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Oxygen saturation (SpO2) 97 % on 24 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Oxygen saturation (SpO2) 98 % on 28 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Systolic blood pressure 92 mmHg on 16 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Systolic blood pressure 103 mmHg on 20 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a

1 · Fever and inflammation with abnormal vital signs (continued)

HIGH PRIORITY MEDICAL REVIEW

Contradicting / limiting (cont.)	resolution. • Systolic blood pressure 112 mmHg on 24 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution. • Systolic blood pressure 117 mmHg on 28 Nov is in a lower NEWS2 band than earlier — a possible improvement, not a resolution.
ALTERNATIVES (P58)	• Intense exercise before sampling. • Sample handling (IL-6 can change with processing delay). • Assay differences between examinations.
MISSING CONTEXT	• Level of consciousness / mentation and supplemental oxygen — needed for NEWS2 and qSOFA. • Symptoms, examination findings and time course. • qSOFA is not recommended as a single screening tool (SSC 2021); it is shown only as context.
CONFIRM	• Ground Medical to assess the crew member clinically: symptoms, mentation and a full repeated set of vital signs at the same time. • If Ground Medical judges it appropriate: repeat CRP and a full blood count, and investigate a possible source of infection according to the clinical findings. • Record supplemental oxygen and consciousness so that an aggregate NEWS2 score can be calculated.
SOURCES · GM STATUS	NEWS2, RCP 2017; CDC; CDC/AHA 2003; Sepsis-3, 2016; SSC 2021; BTS 2017. Ground Medical: Pending Ground Medical review (session review — not persisted).

2 · Raised D-dimer; fibrinogen without an applicable criterion

HIGH PRIORITY MEDICAL REVIEW

WARNING PRIORITY	HIGH PRIORITY MEDICAL REVIEW (rule PP-2). PP-2: D-dimer above the cut-off converges with SpO2 or respiratory rate in a NEWS2 ≥2-point band within one day.
CLINICAL CLASS	• NEUTRAL — no applicable criterion classifies the contributing results; the pattern rests on cited thresholds and review rules, not on a clinical class.
OBSERVED	• D-dimer 0.82 mg/L FEU on 9 Nov — above the conventional 0.5 mg/L FEU cut-off; cut-offs are assay-specific and age-adjusted above 50 years (ESC 2019); age is not recorded. • D-dimer 2.8 mg/L FEU on 12 Nov — above the conventional 0.5 mg/L FEU cut-off; cut-offs are assay-specific and age-adjusted above 50 years (ESC 2019); age is not recorded. • D-dimer 2.1 mg/L FEU on 16 Nov — above the conventional 0.5 mg/L FEU cut-off; cut-offs are assay-specific and age-adjusted above 50 years (ESC 2019); age is not recorded. • D-dimer 1.2 mg/L FEU on 20 Nov — above the conventional 0.5 mg/L FEU cut-off; cut-offs are assay-specific and age-adjusted above 50 years (ESC 2019); age is not recorded. • D-dimer 0.69 mg/L FEU on 24 Nov — above the conventional 0.5 mg/L FEU cut-off; cut-offs are assay-specific and age-adjusted above 50 years (ESC 2019); age is not recorded. • Fibrinogen 295 mg/dL on 3 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 320 mg/dL on 6 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 390 mg/dL on 9 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 520 mg/dL on 12 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 485 mg/dL on 16 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 420 mg/dL on 20 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 350 mg/dL on 24 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Fibrinogen 305 mg/dL on 28 Nov — no criterion is applied; interpret against the performing laboratory's reference interval (not recorded). • Oxygen saturation (SpO2) 91 % on 12 Nov (NEWS2 3-point band). • Oxygen saturation (SpO2) 93 % on 16 Nov (NEWS2 2-point band). • Respiratory rate 24 breaths/min on 12 Nov (NEWS2 2-point band). • Respiratory rate 21 breaths/min on 16 Nov (NEWS2 2-point band).
MEASUREMENT QUALITY	• The D-dimer assay, its own cut-off and the sampling conditions are not recorded.
CLINICAL PATTERN	A raised D-dimer together with low oxygen saturation or fast breathing is a combination clinicians weigh when estimating the probability of venous thromboembolism, including pulmonary embolism. D-dimer and fibrinogen also rise with infection and inflammation (fibrinogen is an acute-phase protein; Gabay 1999), so these results cannot indicate thrombosis on their own (ESC 2019).

2 · Raised D-dimer; fibrinogen without an applicable criterion (continued)

HIGH PRIORITY MEDICAL REVIEW

HYPOTHESES	• Acute-phase response to the inflammation recorded in the same period. • Venous thromboembolism — to be assessed clinically; not indicated by the measurements alone.
CONTRADICTING / LIMITING	• Concurrent inflammation (hsCRP 28 mg/L; hsCRP 22 mg/L; hsCRP 12 mg/L; Body temperature 38.8 °C; Body temperature 38.2 °C) is an alternative reason for the raised D-dimer and for the fibrinogen values.
MISSING CONTEXT	• Clinical probability assessment with a validated score (symptoms, signs, risk factors). • Age (for the age-adjusted D-dimer cut-off) and the assay's own cut-off. • Platelet count — no accepted observation in this period.
CONFIRM	• Ground Medical to estimate the clinical probability of venous thromboembolism with a validated score before deciding on further testing; imaging only if the clinician judges it indicated.
SOURCES · GM STATUS	ESC 2019; Gabay 1999. Ground Medical: Pending Ground Medical review (session review — not persisted).

3 · HbA1c results need verification

DATA CHECK

WARNING PRIORITY	DATA CHECK (rule PP-5). PP-5: every HbA1c result is held by the existing measurement-consistency rule; no clinical statement is made until it is repeated.
CLINICAL CLASS	• NEUTRAL — no applicable criterion classifies the contributing results; the pattern rests on cited thresholds and review rules, not on a clinical class.
OBSERVED	• HbA1c 5.8 % on 3 Nov → 5.9 % on 6 Nov → 9.1 % on 9 Nov → 5.9 % on 12 Nov → 5.8 % on 16 Nov → 5.8 % on 20 Nov → 5.8 % on 24 Nov → 5.8 % on 28 Nov.
MEASUREMENT QUALITY	• HbA1c reflects average glucose over about 2–3 months (ADA Standards of Care), so the change of 3.2 percentage points between 6 Nov and 9 Nov (3 days) cannot be physiological.
CLINICAL PATTERN	These 8 HbA1c results are not mutually consistent; until they are verified, none of them supports or excludes a disorder of blood-sugar regulation.
HYPOTHESES	• Sample mix-up or transcription error. • Different analytical methods. • Altered red-cell turnover affecting HbA1c (haemoglobin 14.5 g/dL → 14.2 g/dL → 13.8 g/dL → 12.6 g/dL → 12.4 g/dL → 12.9 g/dL → 13.5 g/dL → 14.0 g/dL recorded).
ALTERNATIVES (P58)	• Sample mix-up or data-entry error, especially when HbA1c changes markedly within days. • Conditions altering red-cell turnover (anemia, blood loss, transfusion, hemolysis) or hemoglobin variants affecting HbA1c. • A non-fasting or random glucose interpreted as fasting.
MISSING CONTEXT	• HbA1c method and laboratory.
CONFIRM	• Repeat HbA1c on a new sample with the method recorded, together with a fasting plasma glucose, if Ground Medical agrees.
SOURCES · GM STATUS	ADA Standards of Care. Ground Medical: Pending Ground Medical review (session review — not persisted).

4 · Electrolyte shifts with a prolonged QTc

DATA CHECK MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: abnormal value(s) without convergence. A DATA CHECK also applies; it does not lower this priority.
CLINICAL CLASS	• Sodium 128 mmol/L (11/09): AMBER (PROVISIONAL) — Moderate hyponatraemia (biochemical, ESE) • Sodium 131 mmol/L (11/12): AMBER (PROVISIONAL) — Mild hyponatraemia (biochemical, ESE) • Sodium 146 mmol/L (11/20): AMBER (PROVISIONAL) — Hypernatraemia (StatPearls) • Potassium 5.8 mmol/L (11/09): AMBER (PROVISIONAL) — Mild hyperkalaemia (UKKA) • Potassium 3.2 mmol/L (11/20): AMBER (PROVISIONAL) — Mild hypokalaemia (StatPearls)
OBSERVED	• Sodium 139 → 136 → 128 → 131 → 138 → 146 → 141 → 139 mmol/L (3 Nov, 6 Nov, 9 Nov, 12 Nov, 16 Nov, 20 Nov, 24 Nov, 28 Nov): 128 — Moderate hyponatraemia (biochemical, ESE); 131 — Mild hyponatraemia (biochemical, ESE); 146 — Hypernatraemia (StatPearls) (provisional). • Potassium 4.3 → 4.8 → 5.8 → 5.3 → 4.1 → 3.2 → 3.8 → 4.2 mmol/L (3 Nov, 6 Nov, 9 Nov, 12 Nov, 16 Nov, 20 Nov, 24 Nov, 28 Nov): 5.8 — Mild hyperkalaemia (UKKA); 3.2 — Mild hypokalaemia (StatPearls) (provisional). • QTc 510 ms on 12 Nov — above 500 ms, regarded as highly abnormal and associated with a higher risk of torsade de pointes (AHA/ACCF 2010).
MEASUREMENT QUALITY	• High (5.8 mmol/L on 9 Nov) and low (3.2 mmol/L on 20 Nov) potassium results were recorded on different examinations; in-vitro haemolysis or delayed processing can falsely raise an individual result (pseudohyperkalaemia) (StatPearls), and the haemolysis index is not recorded. • The QT correction formula and the heart rate at the time of the ECG are not recorded; Bazett's formula overestimates QTc at faster heart rates, so manual re-measurement matters.

4 · Electrolyte shifts with a prolonged QTc (continued)

DATA CHECK | MONITORING OR FURTHER EVALUATION

CLINICAL PATTERN	Sodium and potassium were both outside the provisional reference range on more than one examination (sodium 9 Nov, 12 Nov, 20 Nov; potassium 9 Nov, 20 Nov), over 11 days. Low potassium (3.2 mmol/L on 20 Nov) and a QTc of 510 ms are both recognised risk factors for torsade de pointes; they were recorded on different days, so their co-existence is not established. If confirmed, these changes matter for heart rhythm and neurological safety; the measurements alone do not establish a cause.
HYPOTHESES	• Pre-analytical or sample-handling error (haemolysis, line contamination, sample mix-up). • A true shift from fluid balance, treatment or illness (treatment and fluid balance are not recorded). • QT-prolonging medicines (medication list not recorded).
CONTRADICTING / LIMITING	• Without kidney-function results the electrolyte changes cannot be placed in context.
ALTERNATIVES (P58)	• Sample hemolysis or delayed processing, which can falsely raise measured potassium. • Contamination from an intravenous line. • Dilution or concentration of the sample. • Laboratory variation between examinations.
MISSING CONTEXT	• Magnesium — no accepted observation in this period. • Creatinine / eGFR (kidney function) — no accepted observation in this period. • Haemolysis index, sample processing time and analyser method. • Fluid intake/output, intravenous therapy and medications. • ECG rhythm, heart rate at recording and QT correction formula.
CONFIRM	• Do not change fluid, salt or potassium intake before Ground Medical review. • If Ground Medical judges it appropriate: repeat sodium and potassium on a new, promptly processed sample with the haemolysis index, and add magnesium and kidney function. • If Ground Medical judges it appropriate: repeat a 12-lead ECG with manual QT measurement and the correction formula recorded.
SOURCES · GM STATUS	Society for Endocrinology 2022; StatPearls; AHA/ACCF 2010. Ground Medical: Pending Ground Medical review (session review — not persisted).

5 · Raised triglycerides with an unfavourable cholesterol profile

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: provisional AMBER lipid values; fasting status is not recorded, so the triglyceride bands cannot support high priority.
CLINICAL CLASS	• Triglycerides 155 mg/dL (11/03): AMBER (PROVISIONAL) — Borderline high (NHLBI) • Triglycerides 165 mg/dL (11/06): AMBER (PROVISIONAL) — Borderline high (NHLBI) • Triglycerides 230 mg/dL (11/09): AMBER (PROVISIONAL) — High (NHLBI) • Triglycerides 330 mg/dL (11/12): AMBER (PROVISIONAL) — High (NHLBI) • Triglycerides 288 mg/dL (11/16): AMBER (PROVISIONAL) — High (NHLBI) • Triglycerides 216 mg/dL (11/20): AMBER (PROVISIONAL) — High (NHLBI) • Triglycerides 184 mg/dL (11/24): AMBER (PROVISIONAL) — Borderline high (NHLBI) • Triglycerides 160 mg/dL (11/28): AMBER (PROVISIONAL) — Borderline high (NHLBI) • HDL cholesterol 39 mg/dL (11/12): AMBER (PROVISIONAL) — Low HDL (ATP III)
OBSERVED	• Triglycerides 155 → 165 → 230 → 330 → 288 → 216 → 184 → 160 mg/dL (3 Nov, 6 Nov, 9 Nov, 12 Nov, 16 Nov, 20 Nov, 24 Nov, 28 Nov), peak 330 on 12 Nov; Borderline high (NHLBI) / High (NHLBI) (provisional; fasting not recorded). • HDL cholesterol 39 mg/dL on 12 Nov — Low HDL (ATP III) (provisional).
MEASUREMENT QUALITY	• Fasting status and time since the last meal are not recorded; triglycerides rise for several hours after eating.
CLINICAL PATTERN	Triglycerides peaked on 12 Nov and fell at the next measurement (288 mg/dL on 16 Nov, 4 days later). A transient peak of this shape fits an acute effect — a recent meal, or the inflammatory response recorded on the same day, which raises plasma triglycerides (Khovidhunkit 2004) — better than a stable lipid disorder. Low HDL and raised LDL, if confirmed when fasting and well, are long-term cardiovascular risk factors.
HYPOTHESES	• Inflammation-related rise in triglycerides (acute-phase lipid changes). • Non-fasting samples.
CONTRADICTING / LIMITING	• The fall after 12 Nov argues against a stable, severe hypertriglyceridaemia.
ALTERNATIVES (P58)	• Non-fasting sample (the most common cause of a single high value). • Recent alcohol intake. • Medications or other metabolic conditions affecting lipids. • Laboratory or sample-handling variation.
MISSING CONTEXT	• Fasting status for each lipid sample. • Total cholesterol and non-HDL cholesterol.
CONFIRM	• Repeat a fasting lipid profile once the acute illness has resolved, if Ground Medical agrees.
SOURCES · GM STATUS	Khovidhunkit 2004. Ground Medical: Pending Ground Medical review (session review — not persisted).

6 · Bone-turnover change and low vitamin D

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: a significant change (trend review) and/or a provisional AMBER vitamin D value; no threshold supports high priority.
CLINICAL CLASS	• Vitamin D (25-hydroxy) 12 ng/mL (11/03): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 12 ng/mL (11/06): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 12 ng/mL (11/09): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 12 ng/mL (11/12): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 13 ng/mL (11/16): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 13 ng/mL (11/20): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 14 ng/mL (11/24): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS) • Vitamin D (25-hydroxy) 14 ng/mL (11/28): AMBER (PROVISIONAL) — Generally considered inadequate (NIH ODS)
OBSERVED	• Osteocalcin 3.5 ng/mL on 3 Nov → 4.1 ng/mL on 6 Nov → 5.2 ng/mL on 9 Nov → 31 ng/mL on 12 Nov → 25 ng/mL on 16 Nov → 18 ng/mL on 20 Nov → 11 ng/mL on 24 Nov → 7.2 ng/mL on 28 Nov (8.9-fold rise to a peak on 12 Nov; last value 7.2 ng/mL on 28 Nov, 2.057× the first). • 25-hydroxyvitamin D 12 ng/mL on 3 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 12 ng/mL on 6 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 12 ng/mL on 9 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 12 ng/mL on 12 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 13 ng/mL on 16 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 13 ng/mL on 20 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 14 ng/mL on 24 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • 25-hydroxyvitamin D 14 ng/mL on 28 Nov — Generally considered inadequate (NIH ODS) (provisional; NIH ODS). • CTX (bone breakdown) 0.32 → 0.35 → 0.41 → 0.73 → 0.66 → 0.55 → 0.45 → 0.36 ng/mL (8 measurements, 3 Nov–28 Nov; range 0.32–0.73) — no criterion applied. • P1NP (bone formation) 43 → 45 → 49 → 68 → 65 → 57 → 51 → 46 ng/mL (8 measurements, 3 Nov–28 Nov; range 43–68) — no criterion applied.
MEASUREMENT QUALITY	• Osteocalcin is unstable after sampling and varies through the day; the IOF-IFCC recommend P1NP and CTX as the reference bone-turnover markers (IOF-IFCC 2011). A change of this size within 25 days should be verified before biological interpretation.
CLINICAL PATTERN	A large, rapid osteocalcin change cannot yet be read as a change in bone formation; the reference markers (P1NP, CTX) were measured repeatedly but have no applicable criterion, so their series are listed for Ground Medical comparison rather than interpreted. Vitamin D in the range generally considered inadequate is relevant to bone health in long-duration spaceflight and should be reviewed by Ground Medical.
HYPOTHESES	• Pre-analytical variation (sample handling, time of day, assay). • Limited sunlight exposure or intake (intake not recorded).
CONTRADICTING / LIMITING	• The reference markers (P1NP, CTX) have no applicable criterion, so their repeated measurements cannot yet confirm or refute the osteocalcin change.
ALTERNATIVES (P58)	• Assay or laboratory differences (osteocalcin is assay-dependent). • Circadian and fasting effects on bone markers. • Recent fracture healing or intense loading.
MISSING CONTEXT	• Calcium, phosphate and parathyroid hormone measured together with the bone markers. • Sampling time (fasting morning samples are recommended for CTX).
CONFIRM	• Repeat bone-turnover markers (P1NP, CTX, osteocalcin) on fasting morning samples together with calcium, phosphate and PTH, if Ground Medical agrees; Ground Medical to review vitamin D adequacy.
SOURCES · GM STATUS	IOF-IFCC 2011; NIH ODS. Ground Medical: Pending Ground Medical review (session review — not persisted).

7 · Poor sleep efficiency and reduced heart-rate variability

MONITORING OR FURTHER EVALUATION

WARNING PRIORITY	MONITORING OR FURTHER EVALUATION (rule PP-4). PP-4: provisional AMBER sleep efficiency and/or a significant HRV change (trend review); no threshold supports high priority.
CLINICAL CLASS	• Sleep efficiency 69 % (11/09): AMBER (PROVISIONAL) — ≤74 % — does not indicate good sleep quality (NSF) • Sleep efficiency 52 % (11/12): AMBER (PROVISIONAL) — ≤74 % — does not indicate good sleep quality (NSF) • Sleep efficiency 56 % (11/16): AMBER (PROVISIONAL) — ≤74 % — does not indicate good sleep quality (NSF) • Sleep efficiency 66 % (11/20): AMBER (PROVISIONAL) — ≤74 % — does not indicate good sleep quality (NSF)
OBSERVED	• Sleep efficiency 69 % on 9 Nov — ≤74 % — does not indicate good sleep quality (NSF) (provisional; NSF 2017). • Sleep efficiency 52 % on 12 Nov — ≤74 % — does not indicate good sleep quality (NSF) (provisional; NSF 2017). • Sleep efficiency 56 % on 16 Nov — ≤74 % — does not indicate good sleep quality (NSF) (provisional; NSF 2017). • Sleep efficiency 66 % on 20 Nov — ≤74 % — does not indicate good sleep quality (NSF) (provisional; NSF 2017). • RMSSD 32 ms on 3 Nov → 29 ms on 6 Nov → 24 ms on 9 Nov → 12 ms on 12 Nov → 14 ms on 16 Nov → 19 ms on 20 Nov → 25 ms on 24 Nov → 30 ms on 28 Nov — significant change (trend review); no criterion classifies HRV.

7 · Poor sleep efficiency and reduced heart-rate variability (continued)

MONITORING OR FURTHER EVALUATION

MEASUREMENT QUALITY	• HRV depends strongly on recording length, posture, breathing and time of day (ESC/NASPE 1996); these conditions are not recorded.
CLINICAL PATTERN	Disturbed sleep and lower parasympathetic (vagal) activity coincide with the acute physiological pattern; they indicate reduced recovery but do not identify a cause.
HYPOTHESES	• Acute illness, fever and stress reducing sleep quality and HRV. • Workload, schedule or environmental disruption (not recorded).
ALTERNATIVES (P58)	• Different measurement posture, time of day or device between examinations. • Recent caffeine, exercise or meals before the recording. • Short recording duration.
MISSING CONTEXT	• Sleep log, workload and schedule; HRV recording conditions.
CONFIRM	• If Ground Medical agrees, repeat HRV under standardised conditions (same time of day, posture and duration) once the acute illness has resolved.
SOURCES · GM STATUS	ESC/NASPE 1996; NSF 2017. Ground Medical: Pending Ground Medical review (session review — not persisted).

Personalised recommendations

Each recommendation is linked to the finding that generated it. GM marks actions that require a Ground Medical decision. No medication or supplement doses are given.

Nutrition

- Record meal timing and content before blood tests. Diets built on vegetables, legumes, whole grains and adequate protein, with fewer refined sugars and sugary drinks, are generally associated with steadier blood glucose; individual changes should follow Ground Medical advice.
  → Possible disturbance of blood-sugar regulation (dysglycemia): HbA1c (3-month average blood sugar) 5.8 % on 2026-11-28; Fasting glucose …
- Triglycerides respond strongly to recent sugars, refined carbohydrates and alcohol; recording intake in the 24 hours before sampling helps interpret the result. Dietary changes should be agreed with Ground Medical.
  → Raised blood triglycerides (possible lipid-metabolism concern): Triglycerides 160 mg/dL on 2026-11-28; LDL cholesterol 125 mg/dL …
- GM Ground Medical should review calcium and vitamin D adequacy; any decision about supplementation belongs to Ground Medical.
  → Change in bone-turnover markers (possible bone-health concern): Osteocalcin (bone formation) 7.2 ng/mL on 2026-11-28; CTX …
- GM Do not change fluid or salt intake on the basis of these results until Ground Medical has reviewed them; the appropriate direction depends on the cause.
  → Possible electrolyte disturbance (sodium / potassium): Sodium 139 mmol/L on 2026-11-28; Potassium 4.2 mmol/L on 2026-11-28

Exercise

- GM Follow Ground Medical's advice on physical activity until the acute findings have been assessed; general exercise suggestions are deferred.
  → Fever and inflammation with abnormal vital signs

Sleep and recovery

- Protect the scheduled sleep period and record sleep duration around the next recordings.
  → Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 30 ms on 2026-11-28
- Record exercise intensity, sleep and any illness symptoms in the days before the next blood test so the inflammatory markers can be placed in context.
  → Possible inflammatory response: hsCRP (inflammation) 2.1 mg/L on 2026-11-28; IL-6 (inflammatory signal) 8.1 pg/mL on 2026-11-28
- Review training load and rest days with the exercise team when heart-rate variability falls.
  → Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 30 ms on 2026-11-28
- Record heart-rate variability under the same posture, time of day and device so that changes can be compared.
  → Reduced heart-rate variability (possible recovery or autonomic strain): Heart-rate variability (RMSSD) 30 ms on 2026-11-28
- Repeat the changed measurements in red blood cells and blood volume with the same laboratory method at the next scheduled examination, so the change can be confirmed.
  → Red blood cells and blood volume (change below the review rules)

Medical follow-up

- GM Ground Medical to assess the crew member clinically: symptoms, mentation and a full repeated set of vital signs at the same time.
  → Fever and inflammation with abnormal vital signs
- GM If Ground Medical judges it appropriate: repeat CRP and a full blood count, and investigate a possible source of infection according to the clinical findings.
  → Fever and inflammation with abnormal vital signs
- GM Ground Medical to estimate the clinical probability of venous thromboembolism with a validated score before deciding on further testing; imaging only if the clinician judges it indicated.
  → Raised D-dimer; fibrinogen without an applicable criterion
- GM Repeat HbA1c on a new sample with the method recorded, together with a fasting plasma glucose, if Ground Medical agrees.
  → HbA1c results need verification
- GM Do not change fluid, salt or potassium intake before Ground Medical review.
  → Electrolyte shifts with a prolonged QTc
- GM Repeat a fasting lipid profile once the acute illness has resolved, if Ground Medical agrees.
  → Raised triglycerides with an unfavourable cholesterol profile
- GM Repeat bone-turnover markers (P1NP, CTX, osteocalcin) on fasting morning samples together with calcium, phosphate and PTH, if Ground Medical agrees; Ground Medical to review vitamin D adequacy.
  → Bone-turnover change and low vitamin D
- GM If Ground Medical agrees, repeat HRV under standardised conditions (same time of day, posture and duration) once the acute illness has resolved.
  → Poor sleep efficiency and reduced heart-rate variability
- GM Check sample collection and handling, fluid intake and urine output, and kidney function (creatinine or eGFR) alongside the repeat measurements.
  → Fluids, electrolytes and kidney function — Potassium (review flag)
- GM Ground Medical should review the cardiovascular change and confirm that posture, timing and method were the same for both measurements.
  → Heart and circulation — Resting heart rate, Respiratory rate and Lactate (review flag)
- GM Ground Medical should review the clotting-marker change together with the venous ultrasound findings.
  → Blood clotting and venous flow — D-dimer and Fibrinogen (review flag)

Complete results — all observations

248 observations in 31 biomarker series from 8 examination date(s), exactly as recorded (no unit conversion). Classes use approved criteria or, where marked PROVISIONAL, literature criteria for this synthetic demonstration. Priority is the medical-review priority of the finding the result belongs to (the same label as on page 1, in the charts and in the assessment); "Associated with HIGH PRIORITY pattern" marks a result whose own class is not high but which belongs to a high-priority pattern; Trend / data shows a significant change between examinations (TREND REVIEW) and observation-level data-quality checks (CHECK) separately.

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
FLUIDS & ELECTROLYTES						
Potassium	**4.3 mmol/L**	11/03	NEUTRAL	MONITORING + pattern check †	TREND REVIEW	— †
Potassium	**4.8 mmol/L**	11/06	NEUTRAL	MONITORING + pattern check †	TREND REVIEW	— †

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Potassium	5.8 mmol/L	11/09	AMBER PROV.	MONITORING + pattern check †	TREND REVIEW	Mild hyperkalaemia (UKKA) (5.5–5.9 mmol/L) ‡ †
Potassium	5.3 mmol/L	11/12	NEUTRAL	MONITORING + pattern check †	TREND REVIEW	— †
Potassium	4.1 mmol/L	11/16	NEUTRAL	MONITORING + pattern check †	TREND REVIEW	— †
Potassium	3.2 mmol/L	11/20	AMBER PROV.	MONITORING + pattern check †	TREND REVIEW	Mild hypokalaemia (StatPearls) (≥3 and <3.5 mmol/L) †
Potassium	3.8 mmol/L	11/24	NEUTRAL	MONITORING + pattern check †	TREND REVIEW	— †
Potassium	4.2 mmol/L	11/28	NEUTRAL	MONITORING + pattern check †	TREND REVIEW	— †
Sodium	139 mmol/L	11/03	NEUTRAL	MONITORING	—	—
Sodium	136 mmol/L	11/06	NEUTRAL	MONITORING	—	—
Sodium	128 mmol/L	11/09	AMBER PROV.	MONITORING	—	Moderate hyponatraemia (biochemical, ESE) (125–129 mmol/L) ‡
Sodium	131 mmol/L	11/12	AMBER PROV.	MONITORING	—	Mild hyponatraemia (biochemical, ESE) (≥130 and <135 mmol/L) ‡
Sodium	138 mmol/L	11/16	NEUTRAL	MONITORING	—	—
Sodium	146 mmol/L	11/20	AMBER PROV.	MONITORING	—	Hypernatraemia (StatPearls) (>145 mmol/L)
Sodium	141 mmol/L	11/24	NEUTRAL	MONITORING	—	—
Sodium	139 mmol/L	11/28	NEUTRAL	MONITORING	—	—
GLUCOSE & LIPIDS						
Fasting glucose	94 mg/dL	11/03	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	108 mg/dL	11/06	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	136 mg/dL	11/09	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	172 mg/dL	11/12	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	158 mg/dL	11/16	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	119 mg/dL	11/20	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	103 mg/dL	11/24	NEUTRAL	—	TREND REVIEW	— ◊
Fasting glucose	96 mg/dL	11/28	NEUTRAL	—	TREND REVIEW	— ◊
HbA1c (3-month average blood sugar)	5.8 %	11/03	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	5.9 %	11/06	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	9.1 %	11/09	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	5.9 %	11/12	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	5.8 %	11/16	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	5.8 %	11/20	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	5.8 %	11/24	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HbA1c (3-month average blood sugar)	5.8 %	11/28	NEUTRAL	DATA CHECK	TREND REVIEW CHECK	Held — verify the measurement first
HDL cholesterol	48 mg/dL	11/03	NEUTRAL	MONITORING	—	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
HDL cholesterol	**46 mg/dL**	11/06	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**43 mg/dL**	11/09	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**39 mg/dL**	11/12	AMBER PROV.	MONITORING	—	Low HDL (ATP III) (<40 mg/dL) § ◊
HDL cholesterol	**40 mg/dL**	11/16	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**42 mg/dL**	11/20	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**44 mg/dL**	11/24	NEUTRAL	MONITORING	—	— ◊
HDL cholesterol	**46 mg/dL**	11/28	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**122 mg/dL**	11/03	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**127 mg/dL**	11/06	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**134 mg/dL**	11/09	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**145 mg/dL**	11/12	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**143 mg/dL**	11/16	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**137 mg/dL**	11/20	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**131 mg/dL**	11/24	NEUTRAL	MONITORING	—	— ◊
LDL cholesterol	**125 mg/dL**	11/28	NEUTRAL	MONITORING	—	— ◊
Triglycerides	**155 mg/dL**	11/03	AMBER PROV.	MONITORING	TREND REVIEW	Borderline high (NHLBI) (150–199 mg/dL) ‡
Triglycerides	**165 mg/dL**	11/06	AMBER PROV.	MONITORING	TREND REVIEW	Borderline high (NHLBI) (150–199 mg/dL) ‡
Triglycerides	**230 mg/dL**	11/09	AMBER PROV.	MONITORING	TREND REVIEW	High (NHLBI) (200–499 mg/dL) ‡
Triglycerides	**330 mg/dL**	11/12	AMBER PROV.	MONITORING	TREND REVIEW	High (NHLBI) (200–499 mg/dL) ‡
Triglycerides	**288 mg/dL**	11/16	AMBER PROV.	MONITORING	TREND REVIEW	High (NHLBI) (200–499 mg/dL) ‡
Triglycerides	**216 mg/dL**	11/20	AMBER PROV.	MONITORING	TREND REVIEW	High (NHLBI) (200–499 mg/dL) ‡
Triglycerides	**184 mg/dL**	11/24	AMBER PROV.	MONITORING	TREND REVIEW	Borderline high (NHLBI) (150–199 mg/dL) ‡
Triglycerides	**160 mg/dL**	11/28	AMBER PROV.	MONITORING	TREND REVIEW	Borderline high (NHLBI) (150–199 mg/dL) ‡
INFLAMMATION & IMMUNITY						
hsCRP (inflammation)	**1.2 mg/L**	11/03	NEUTRAL	—	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**2.8 mg/L**	11/06	NEUTRAL	—	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**8.4 mg/L**	11/09	NEUTRAL	—	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**28 mg/L**	11/12	AMBER PROV.	Associated with HIGH PRIORITY pattern	TREND REVIEW	Above 10 mg/L — repeat and evaluate (CDC/AHA) (>10 mg/L)
hsCRP (inflammation)	**22 mg/L**	11/16	AMBER PROV.	Associated with HIGH PRIORITY pattern	TREND REVIEW	Above 10 mg/L — repeat and evaluate (CDC/AHA) (>10 mg/L)
hsCRP (inflammation)	**12 mg/L**	11/20	AMBER PROV.	Associated with HIGH PRIORITY pattern	TREND REVIEW	Above 10 mg/L — repeat and evaluate (CDC/AHA) (>10 mg/L)
hsCRP (inflammation)	**5.2 mg/L**	11/24	NEUTRAL	—	TREND REVIEW	Candidate criterion pending approval
hsCRP (inflammation)	**2.1 mg/L**	11/28	NEUTRAL	—	TREND REVIEW	Candidate criterion pending approval
IL-10 (anti-inflammatory signal)	**4 pg/mL**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**5 pg/mL**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**8 pg/mL**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**15 pg/mL**	11/12	NEUTRAL	—	TREND REVIEW	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
IL-10 (anti-inflammatory signal)	**13 pg/mL**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**9 pg/mL**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**6 pg/mL**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
IL-10 (anti-inflammatory signal)	**5 pg/mL**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
IL-6 (inflammatory signal)	**5.8 pg/mL**	11/03	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**12 pg/mL**	11/06	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**32 pg/mL**	11/09	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**74 pg/mL**	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**55 pg/mL**	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**29 pg/mL**	11/20	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**15 pg/mL**	11/24	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
IL-6 (inflammatory signal)	**8.1 pg/mL**	11/28	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
TNF-α	**8 pg/mL**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**10 pg/mL**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**15 pg/mL**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**24 pg/mL**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**20 pg/mL**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**15 pg/mL**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**11 pg/mL**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
TNF-α	**9 pg/mL**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
BLOOD & CLOTTING						
aPTT (clotting time)	**29 s**	11/03	NEUTRAL	—	—	—
aPTT (clotting time)	**30 s**	11/06	NEUTRAL	—	—	—
aPTT (clotting time)	**31 s**	11/09	NEUTRAL	—	—	—
aPTT (clotting time)	**34 s**	11/12	NEUTRAL	—	—	—
aPTT (clotting time)	**32 s**	11/16	NEUTRAL	—	—	—
aPTT (clotting time)	**31 s**	11/20	NEUTRAL	—	—	—
aPTT (clotting time)	**30 s**	11/24	NEUTRAL	—	—	—
aPTT (clotting time)	**29 s**	11/28	NEUTRAL	—	—	—
D-dimer	**0.32 mg/L FEU**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**0.41 mg/L FEU**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
D-dimer	**0.82 mg/L FEU**	11/09	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
D-dimer	**2.8 mg/L FEU**	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
D-dimer	**2.1 mg/L FEU**	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
D-dimer	**1.2 mg/L FEU**	11/20	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
D-dimer	**0.69 mg/L FEU**	11/24	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
D-dimer	**0.39 mg/L FEU**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
Fibrinogen	**295 mg/dL**	11/03	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**320 mg/dL**	11/06	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**390 mg/dL**	11/09	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**520 mg/dL**	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**485 mg/dL**	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**420 mg/dL**	11/20	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**350 mg/dL**	11/24	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Fibrinogen	**305 mg/dL**	11/28	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Hematocrit	**43 %**	11/03	NEUTRAL	—	—	—
Hematocrit	**42 %**	11/06	NEUTRAL	—	—	—
Hematocrit	**41 %**	11/09	NEUTRAL	—	—	—
Hematocrit	**38 %**	11/12	NEUTRAL	—	—	—
Hematocrit	**37 %**	11/16	NEUTRAL	—	—	—
Hematocrit	**39 %**	11/20	NEUTRAL	—	—	—
Hematocrit	**40 %**	11/24	NEUTRAL	—	—	—
Hematocrit	**42 %**	11/28	NEUTRAL	—	—	—
Hemoglobin	**14.5 g/dL**	11/03	NEUTRAL	—	—	—
Hemoglobin	**14.2 g/dL**	11/06	NEUTRAL	—	—	—
Hemoglobin	**13.8 g/dL**	11/09	NEUTRAL	—	—	—
Hemoglobin	**12.6 g/dL**	11/12	NEUTRAL	—	—	—
Hemoglobin	**12.4 g/dL**	11/16	NEUTRAL	—	—	—
Hemoglobin	**12.9 g/dL**	11/20	NEUTRAL	—	—	—
Hemoglobin	**13.5 g/dL**	11/24	NEUTRAL	—	—	—
Hemoglobin	**14.0 g/dL**	11/28	NEUTRAL	—	—	—
HEART & CIRCULATION						
Diastolic blood pressure	**76 mmHg**	11/03	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**73 mmHg**	11/06	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**69 mmHg**	11/09	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**54 mmHg**	11/12	AMBER PROV.	—	—	Below 60 mmHg (NHLBI low blood pressure) (<60 mmHg) § ◊
Diastolic blood pressure	**58 mmHg**	11/16	AMBER PROV.	—	—	Below 60 mmHg (NHLBI low blood pressure) (<60 mmHg) § ◊
Diastolic blood pressure	**65 mmHg**	11/20	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**72 mmHg**	11/24	NEUTRAL	—	—	— ◊
Diastolic blood pressure	**75 mmHg**	11/28	NEUTRAL	—	—	— ◊
Lactate	**1.1 mmol/L**	11/03	NEUTRAL	—	TREND REVIEW	—

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Lactate	**1.3 mmol/L**	11/06	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.8 mmol/L**	11/09	NEUTRAL	—	TREND REVIEW	—
Lactate	**4.2 mmol/L**	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
Lactate	**3.0 mmol/L**	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	—
Lactate	**1.9 mmol/L**	11/20	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.4 mmol/L**	11/24	NEUTRAL	—	TREND REVIEW	—
Lactate	**1.2 mmol/L**	11/28	NEUTRAL	—	TREND REVIEW	—
QTc interval	**416 ms**	11/03	NEUTRAL	—	—	—
QTc interval	**425 ms**	11/06	NEUTRAL	—	—	—
QTc interval	**447 ms**	11/09	NEUTRAL	—	—	—
QTc interval	**510 ms**	11/12	NEUTRAL	MONITORING	—	—
QTc interval	**484 ms**	11/16	NEUTRAL	—	—	—
QTc interval	**455 ms**	11/20	NEUTRAL	—	—	—
QTc interval	**433 ms**	11/24	NEUTRAL	—	—	—
QTc interval	**421 ms**	11/28	NEUTRAL	—	—	—
Resting heart rate	**64 bpm**	11/03	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**69 bpm**	11/06	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**77 bpm**	11/09	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**108 bpm**	11/12	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**98 bpm**	11/16	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**83 bpm**	11/20	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**72 bpm**	11/24	NEUTRAL	—	TREND REVIEW	— ◊
Resting heart rate	**66 bpm**	11/28	NEUTRAL	—	TREND REVIEW	— ◊
Systolic blood pressure	**118 mmHg**	11/03	NEUTRAL	—	—	— ◊
Systolic blood pressure	**114 mmHg**	11/06	NEUTRAL	—	—	— ◊
Systolic blood pressure	**108 mmHg**	11/09	NEUTRAL	—	—	— ◊
Systolic blood pressure	**88 mmHg**	11/12	AMBER PROV.	Associated with HIGH PRIORITY pattern	—	Below 90 mmHg (NHLBI low blood pressure) (<90 mmHg) § ◊
Systolic blood pressure	**92 mmHg**	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	—	— ◊
Systolic blood pressure	**103 mmHg**	11/20	NEUTRAL	—	—	— ◊
Systolic blood pressure	**112 mmHg**	11/24	NEUTRAL	—	—	— ◊
Systolic blood pressure	**117 mmHg**	11/28	NEUTRAL	—	—	— ◊
BONE & MINERAL						
CTX (bone breakdown)	**0.32 ng/mL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.35 ng/mL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.41 ng/mL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.73 ng/mL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.66 ng/mL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.55 ng/mL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.45 ng/mL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	— ◊
CTX (bone breakdown)	**0.36 ng/mL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**3.5 ng/mL**	11/03	NEUTRAL	MONITORING	TREND REVIEW	— ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Osteocalcin (bone formation)	**4.1 ng/mL**	11/06	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**5.2 ng/mL**	11/09	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**31 ng/mL**	11/12	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**25 ng/mL**	11/16	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**18 ng/mL**	11/20	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**11 ng/mL**	11/24	NEUTRAL	MONITORING	TREND REVIEW	— ◊
Osteocalcin (bone formation)	**7.2 ng/mL**	11/28	NEUTRAL	MONITORING	TREND REVIEW	— ◊
P1NP (bone formation)	**43 ng/mL**	11/03	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**45 ng/mL**	11/06	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**49 ng/mL**	11/09	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**68 ng/mL**	11/12	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**65 ng/mL**	11/16	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**57 ng/mL**	11/20	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**51 ng/mL**	11/24	NEUTRAL	MONITORING	—	— ◊
P1NP (bone formation)	**46 ng/mL**	11/28	NEUTRAL	MONITORING	—	— ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/03	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/06	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/09	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**12 ng/mL**	11/12	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**13 ng/mL**	11/16	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**13 ng/mL**	11/20	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**14 ng/mL**	11/24	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
Vitamin D (25-hydroxy)	**14 ng/mL**	11/28	AMBER PROV.	MONITORING	—	Generally considered inadequate (NIH ODS) (≥12 and <20 ng/mL) ‡ § ◊
SLEEP & RECOVERY						
Heart-rate variability (RMSSD)	**32 ms**	11/03	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**29 ms**	11/06	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**24 ms**	11/09	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**12 ms**	11/12	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**14 ms**	11/16	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**19 ms**	11/20	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**25 ms**	11/24	NEUTRAL	MONITORING	TREND REVIEW	—
Heart-rate variability (RMSSD)	**30 ms**	11/28	NEUTRAL	MONITORING	TREND REVIEW	—
Sleep efficiency	**83 %**	11/03	NEUTRAL	—	—	— ◊
Sleep efficiency	**78 %**	11/06	NEUTRAL	—	—	— ◊
Sleep efficiency	**69 %**	11/09	AMBER PROV.	MONITORING	—	≤74 % — does not indicate good sleep quality (NSF) (≤74 %) § ◊

Measurement	Value	Date	Class	Priority	Trend / data	Criterion
Sleep efficiency	52 %	11/12	AMBER PROV.	MONITORING	—	≤74 % — does not indicate good sleep quality (NSF) (≤74 %) § ◊
Sleep efficiency	56 %	11/16	AMBER PROV.	MONITORING	—	≤74 % — does not indicate good sleep quality (NSF) (≤74 %) § ◊
Sleep efficiency	66 %	11/20	AMBER PROV.	MONITORING	—	≤74 % — does not indicate good sleep quality (NSF) (≤74 %) § ◊
Sleep efficiency	75 %	11/24	NEUTRAL	—	—	— ◊
Sleep efficiency	81 %	11/28	NEUTRAL	—	—	— ◊
OTHER MEASUREMENTS						
Body temperature	36.7 °C	11/03	NEUTRAL	—	—	— ◊
Body temperature	37.0 °C	11/06	NEUTRAL	—	—	— ◊
Body temperature	37.5 °C	11/09	NEUTRAL	—	—	— ◊
Body temperature	38.8 °C	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	—	— ◊
Body temperature	38.2 °C	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	—	— ◊
Body temperature	37.6 °C	11/20	NEUTRAL	—	—	— ◊
Body temperature	37.1 °C	11/24	NEUTRAL	—	—	— ◊
Body temperature	36.8 °C	11/28	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	98 %	11/03	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	97 %	11/06	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	96 %	11/09	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	91 %	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	—	— ◊
Oxygen saturation (SpO2)	93 %	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	—	— ◊
Oxygen saturation (SpO2)	96 %	11/20	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	97 %	11/24	NEUTRAL	—	—	— ◊
Oxygen saturation (SpO2)	98 %	11/28	NEUTRAL	—	—	— ◊
Respiratory rate	14 breaths/min	11/03	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	15 breaths/min	11/06	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	17 breaths/min	11/09	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	24 breaths/min	11/12	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Respiratory rate	21 breaths/min	11/16	NEUTRAL	Associated with HIGH PRIORITY pattern	TREND REVIEW	— ◊
Respiratory rate	18 breaths/min	11/20	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	16 breaths/min	11/24	NEUTRAL	—	TREND REVIEW	— ◊
Respiratory rate	14 breaths/min	11/28	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	6.1 10^9/L	11/03	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	6.9 10^9/L	11/06	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	8.4 10^9/L	11/09	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	12.8 10^9/L	11/12	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	11.7 10^9/L	11/16	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	9.4 10^9/L	11/20	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	7.6 10^9/L	11/24	NEUTRAL	—	TREND REVIEW	— ◊
White blood cell count	6.5 10^9/L	11/28	NEUTRAL	—	TREND REVIEW	— ◊

— No applicable criterion: NEUTRAL (insufficient classification evidence).

‡ Per-measurement conditions of the provisional criterion that are not recorded: Haemolysis, prolonged transit and delayed separation are recorded as excluded for this sample; The specimen type (serum or lithium-heparin plasma) is recorded for this sample; The observation records that sodium was measured by ion-specific electrode; Hyperglycaemia is recorded as excluded for this sample (or a glucose-corrected sodium is recorded); The observation records the fasting state (fasting time) of the sample; The observation records the 25(OH)D assay method.

§ Identity bound through an owner-directed synthetic demonstration mapping (not Researcher-approved).

◊ Synthetic demonstration catalog identity (not scientifically validated).

† Pattern-level DATA CHECK (Electrolyte shifts with a prolonged QTc): verify this series before interpretation. It is not an observation-level measurement error and does not lower the medical-review priority.

Methodology note

• Data source and census. Census p60-observation-census.v1: 248 observations, 31 series, dates 2026-11-03, 2026-11-06, 2026-11-09, 2026-11-12, 2026-11-16, 2026-11-20, 2026-11-24, 2026-11-28; SHA-256 8aedfd555c01535527e1e1e60cee334b5b7b7f4e5071d281255677e314a0be59. Records held only on another device or only in the cloud are not included unless imported through the explicit cross-device sync. Generator p60-premium-report.v1 · Evidence ID MITO-EV-4035-8855-9E47-82C7.

• Data. 248 observations · 31 biomarker series (exact catalog identity + unit) · 8 examination date(s) · 31 series compared between examinations. Only accepted, stored observations are evaluated; rows held at import for Researcher review (unresolved names or units) are not stored and are not evaluated in this report. Values exactly as recorded; no unit conversion or interpolation. Synthetic demonstration data.

• Calculations. Changes only between consecutive observations of the same identity and unit; the full trajectory, including intermediate peaks and troughs, is reported.

• Criteria. 0 Researcher-approved criteria available. LITERATURE-BASED SYNTHETIC DEMONSTRATION — NOT CLINICALLY VALIDATED: provisional literature interpretation ON (synthetic demonstration only; marked PROVISIONAL; never approved classifications or validated alerts). General-population criteria are not NASA flight limits.

• Synthetic identity mappings. 25 owner-directed mappings re-verified against the repository (same concept, identical unit, same domain, unique target, named mission); not Researcher-approved and never used for approved classification, real data or unit conversion. Used in this report: HDL cholesterol (demo:ubt-5ad8b852 → cv-087, mg/dL); Diastolic blood pressure (demo:ubt-5d16fcaf → cv-006, mmHg); Systolic blood pressure (demo:ubt-e1140df3 → cv-005, mmHg); Vitamin D (25-hydroxy) (demo:ubt-7b25ce69 → criterion:p59-nihods-25ohd, ng/mL); Sleep efficiency (demo:ubt-efe7e2fb → criterion:p59-nsf2017-sleep-efficiency, %).

• Warning priority (panel colour) is a separate axis from clinical class. Every P58 concern in this report is represented by a P60 pattern, so the P59 rules WP-1 to WP-5 are not shown here; they remain in force for concerns that no pattern covers.

• Cross-domain pattern analysis (P60, owner directed synthetic demonstration pending researcher review): 7 pattern(s) from 35 concepts bound by exact identity and unit; priorities by rules PP-1, PP-2, PP-3, PP-4, PP-5. Patterns never change a result's clinical class. Priority rules: PP-1 any vital sign in a NEWS2 3-point band; PP-2 two independent findings converging on the same or adjacent dates; PP-3 sodium change above 10 mmol/L within one day with abnormal values at both ends (plus DATA CHECK); PP-4 single threshold or review change (monitoring); PP-5 all contributing results held (data check).

• Automated assessment. Deterministic clinical reasoning engine. Review priority and data quality are separate from clinical classification. Not a diagnosis; no medication or dose advice.

References for the cross-domain patterns

NEWS2, RCP 2017 — Royal College of Physicians. National Early Warning Score (NEWS) 2: Standardising the assessment of acute-illness severity in the NHS. Updated report of a working party. London: RCP, 2017.

CDC — Centers for Disease Control and Prevention. Definitions of Signs, Symptoms, and Conditions of Ill Travelers (Port Health).

CDC/AHA 2003 — Pearson TA, Mensah GA, Alexander RW, et al. Markers of inflammation and cardiovascular disease: application to clinical and public health practice. A statement for healthcare professionals from the CDC and the AHA. Circulation. 2003;107(3):499-511.

Sepsis-3, 2016 — Singer M, Deutschman CS, Seymour CW, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA. 2016;315(8):801-810.

SSC 2021 — Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. Crit Care Med. 2021;49(11):e1063-e1143.

BTS 2017 — O'Driscoll BR, Howard LS, Earis J, Mak V. BTS guideline for oxygen use in adults in healthcare and emergency settings. Thorax. 2017;72(Suppl 1):ii1-ii90.

ESC 2019 — Konstantinides SV, Meyer G, Becattini C, et al. 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society. Eur Heart J. 2020;41(4):543-603.

Gabay 1999 — Gabay C, Kushner I. Acute-phase proteins and other systemic responses to inflammation. N Engl J Med. 1999;340(6):448-454.

ADA Standards of Care — American Diabetes Association Professional Practice Committee. Glycemic Goals and Hypoglycemia: Standards of Care in Diabetes (section 6).

Society for Endocrinology 2022 — Society for Endocrinology Endocrine Emergency Guidance: Emergency management of severe and moderately severely symptomatic hyponatraemia in adult patients (2022).

StatPearls — Simon LV, Hashmi MF, Farrell MW. Hyperkalemia. StatPearls. Treasure Island (FL): StatPearls Publishing.

AHA/ACCF 2010 — Drew BJ, Ackerman MJ, Funk M, et al. Prevention of torsade de pointes in hospital settings: a scientific statement from the American Heart Association and the American College of Cardiology Foundation. Circulation. 2010;121:1047-1060.

Khovidhunkit 2004 — Khovidhunkit W, Kim MS, Memon RA, et al. Effects of infection and inflammation on lipid and lipoprotein metabolism: mechanisms and consequences to the host. J Lipid Res. 2004;45(7):1169-1196.

IOF-IFCC 2011 — Vasikaran S, Eastell R, Bruyère O, et al. Markers of bone turnover for the prediction of fracture risk and monitoring of osteoporosis treatment: a need for international reference standards. Osteoporos Int. 2011;22(2):391-420.

NIH ODS — NIH Office of Dietary Supplements. Vitamin D — Fact Sheet for Health Professionals.

ESC/NASPE 1996 — Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. Heart rate variability: standards of measurement, physiological interpretation and clinical use. Circulation. 1996;93(5):1043-1065.

NSF 2017 — Ohayon M, Wickwire EM, Hirshkowitz M, et al. National Sleep Foundation's sleep quality recommendations: first report. Sleep Health. 2017;3(1):6-19.